Risky Insurance: Life-Cycle Insurance Portfolio Choice with Incomplete Markets

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We study consumer demand for savings, life insurance, annuities, and long-term care insurance using new survey data and a structural life-cycle model. Individuals perceive substantial nonpayment risk across all three insurance products, and these beliefs predict ownership. Embedding the measured beliefs into an incomplete-markets model alongside additional real-world insurance features, we match empirical patterns of low participation. Relative to a no-insurance benchmark, access to existing imperfect insurance reduces median wealth by 16% and generates a modest 0.6% welfare gain. Eliminating nonpayment risk substantially increases ownership, reduces median wealth by 27%, and raises the welfare gain to 2.3%.

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1 Introduction

Americans have access to a range of financial products to insure against major life-cycle health, longevity, and mortality risks. Three primary products are: (i) life insurance, which insures against premature mortality by increasing bequests to surviving beneficiaries; (ii) annuities, which insure against outliving one's wealth by providing a guaranteed income stream for the duration of life; and (iii) long-term care insurance, which covers costs related to assistance with the activities of daily living, e.g., nursing homes.

Despite the severity and frequency of these risks, only 3% of individuals aged 60–65 own a private annuity, 60% own life insurance, and 12% own long-term care insurance (LTCI). Why individuals choose not to insure against these risks, instead partially self-insuring through savings, remains a central question in household finance and macroeconomics. Low ownership may reflect limited intrinsic demand for resources in these states or crowding out by public insurance. Alternatively, individuals may desire more insurance but find the products available in the private market undesirable.

We depart from the literature, which typically models individual idealized products, by evaluating joint portfolio choice over savings and insurance with realistic features. We make two key contributions. First, using a purpose-designed survey, we document that individuals view insurance itself as inherently risky, harboring substantial concerns that policies will not fully pay out. Second, we embed these empirical nonpayment beliefs—alongside other insurance market frictions—into an incomplete-markets life-cycle model to quantify how these frictions determine portfolio choices, saving, and consumer welfare.

On average, individuals expect to receive 76%–87% of promised payouts conditional on a qualifying event, and perceive annual full-default probabilities ranging from 1.3% to 1.8% across products. Life insurance is consistently viewed as the safest and LTCI as the riskiest. Validating these measures, we find that elicited beliefs predict actual insurance ownership. Model counterfactuals reveal that nonpayment risk beliefs and high pricing loads severely depress insurance demand, encourage precautionary saving, and reduce consumer welfare. Baseline access to existing, imperfect insurance products generates a modest welfare gain of 0.6% of lifetime consumption and reduces median wealth by 16%. Making insurance risk-free substantially increases insurance participation rates—from 55% to 73% for life insurance, 13% to 43% for LTCI, and 2% to 8% for annuities. Consequently, the welfare gain from these idealized insurance products rises to 2.3% and the reduction in median wealth to 27%. Finally, we decompose the sources of these effects and document how public insurance programs like Medicaid and Social Security affect private insurance demand and welfare.

Measured Nonpayment Beliefs. Conditional on a qualifying event, individuals expect to receive only 87.1% of the promised payout for life insurance, 81.5% for annuities, and 76.2% for LTC insurance. Furthermore, their elicited payout distributions reveal substantial uncertainty around these averages. We additionally measure the annual full-default probability. Full default is the event that an insurer permanently ceases all future payments on the policy. On average, respondents believe these full-default probabilities are 1.3% for life insurance, 1.4% for annuities, and 1.8% for LTCI. Auxiliary survey measures reveal that this perceived risk includes not only concerns over insurer solvency, but also broader fears about contract disputes, delays, and procedural complexity in claims processes.

In support of these measures, we show that subjective nonpayment risk predicts actual insurance ownership. Standard demographic controls and behavioral measures predict insurance demand poorly, but our elicited risk measures robustly predict non-participation.

Structural Model Details: Insurance Ownership, Savings, and Welfare. We embed multiple insurance products into an otherwise standard heterogeneous-agent incomplete-markets life-cycle model of consumption and saving. Individuals face health and mortality risk and borrowing constraints. The model incorporates realistic insurance market features, including subjective nonpayment risk, pricing loads, illiquidity (sale discounts), and quantity restrictions such as age cutoffs and short-sale limits. The model also features means-tested public long-term care support (Medicaid) and public annuities (Social Security).

We calibrate the model to match empirical moments of the joint distribution of age, sex, income, wealth, health, mortality, and average insurance ownership rates, alongside real-world insurance product characteristics. In addition, the calibrated model replicates untargeted life-cycle patterns of insurance ownership, with persistently low participation in annuities and long-term care insurance, and high, but age-declining, participation in life insurance. It also broadly matches the correlation of ownership across products.

With the quantified model, we show that product-market frictions play a central role in shaping insurance participation. Eliminating either perceived nonpayment risk or pricing loads yields large increases in ownership. For example, making insurance risk-free raises LTCI participation from 13% to 43%, while removing both frictions pushes it above 50%, with life insurance exhibiting similarly large responses. As individuals substitute from self-insurance toward state-contingent market insurance, liquid wealth falls and savings are drawn down more aggressively in old age. These results on ownership do not depend on whether nonpayment beliefs are rational, since behavior depends only on the beliefs themselves.

We then turn to consumer welfare analysis, where the distinction between beliefs and the true probability of realized outcomes becomes important. Because the available evidence,

reviewed in Section 3.4, does not establish whether these beliefs are accurate, we consider two cases, always holding subjective beliefs fixed at their measured values: (1) beliefs are rational, meaning nonpayment risk is realized as perceived, and (2) insurers always honor their contracts, so beliefs are too pessimistic. Under rational beliefs, the baseline welfare gain from the existence of private insurance is modest at 0.6% of lifetime consumption on average. However, eliminating nonpayment risk raises the gain to 2.3%, and removing both nonpayment risk and price loads increases the gain to 4.1%. In case (2) with non-rational pessimistic beliefs, the baseline welfare gain is mechanically higher at 1.7%—because individuals always receive insurance payments they are due—but remains below the 2.3% achievable if they correctly perceived insurance as risk-free and accordingly purchased more insurance.

Decomposing these welfare gains reveals that LTCI accounts for over half of the total welfare gain from all products, despite substantially greater ownership of life insurance. Furthermore, the risk of annual partial nonpayment suppresses participation and welfare significantly more than the risk of full default, especially for LTCI and life insurance, which together deliver roughly 90% of total welfare gains.

Finally, we demonstrate the central role of public insurance. Means-tested public long-term care (Medicaid) substantially crowds out private LTCI. Reducing the generosity of this safety net spurs private insurance demand but imposes severe, regressive welfare losses. Furthermore, replacing Social Security's public annuity with an upfront lump-sum wealth transfer does not spur private annuitization. Instead, individuals use the liquidity to self-insure. Because they cannot borrow against future benefits, the lump sum relaxes their borrowing constraint and lets them better smooth consumption. It also reduces their life insurance demand, since they have more wealth to bequeath in the event of an early death, when the bequest motive is strongest. Welfare improves on average, even though the private market offers only an imperfect substitute for the public annuity they lose.

In summary, the phenomenon of risky insurance plays a crucial role in determining insurance ownership, life-cycle savings, and the welfare effects of insurance. Eliminating insurance nonpayment risk—whether actual or incorrectly perceived—yields substantial welfare gains by allowing individuals to insure major life-cycle health and mortality risks. Consequently, the welfare cost of not owning risky insurance is much lower than the cost of not owning idealized products that complete markets.

In the remainder of the paper, Section 2 introduces the data, Section 3 analyzes the elicited distributions of insurance nonpayment risk, Section 4 presents the structural model and its calibration, Section 5 presents model results, and Section 6 concludes.

1.1 Relation to the Literature

Life-Cycle Savings. We contribute to the literature on life-cycle saving, which emphasizes that mortality risk, medical expense risk, long-term care risk, and bequest motives are primary drivers of wealth accumulation. It is well established that the desire to leave a bequest affects life-cycle saving (De Nardi, 2004; Lockwood, 2012).[1] It is also well understood that late-in-life medical costs and long-term-care risk affect life-cycle saving (De Nardi, French and Jones, 2010; Ameriks, Briggs, Caplin, Shapiro and Tonetti, 2020).[2]

Models in this literature often treat late-in-life health shocks as privately uninsurable, restricting the asset space to risk-free bonds and forcing elderly households to self-insure through precautionary saving or rely on government care. We build on this framework by introducing a portfolio choice over the private insurance contracts designed to cover these risks. Our contribution is to incorporate insurance portfolio choice and to measure and model nonpayment risk.[3] Consequently, we show that access to private insurance reduces precautionary saving, even though frictions in these markets generate a reliance on self-insurance that would be absent with complete markets.

Insurance Demand and Nonpayment Risk. A large literature examines the demand for annuities (e.g., Yaari (1965); Brown (2001)), life insurance (e.g., Bernheim (1991); Hong and Ríos-Rull (2012)), and LTC insurance (e.g., Brown and Finkelstein (2007, 2008, 2011)). We contribute to the study of the demand for life-cycle insurance products by measuring and modeling imperfect insurance, with an emphasis on the risk of insurance nonpayment.

As discussed by Mayers and Smith (1983) and emphasized in Gomes, Haliassos and Ramadorai (2021), the demand for insurance products should be studied jointly due to the covariance in their payouts.[4] A very useful benchmark for our results is provided by Koijen, Van Nieuwerburgh and Yogo (2016), who evaluate the joint demand for annuities, life insurance, and LTC insurance in a complete-markets framework. We complement this benchmark by characterizing joint insurance demand in a quantitative incomplete-markets model with nonpayment risk.

In addition to incorporating insurance portfolio choice, we connect the life-cycle savings

[1] See also Hurd (1989); Kopczuk and Lupton (2007); Love, Palumbo and Smith (2009).

[2] See also Hubbard, Skinner and Zeldes (1994); Palumbo (1999); Anderson, French and Lam (2004); Kopecky and Koreshkova (2014); Braun, Kopecky and Koreshkova (2017); Lockwood (2012, 2018).

[3] For tractability and computational feasibility, we abstract from other features recently explored in frontier models, such as a more general modeling of family and labor supply (De Nardi, French, Jones and McGee, 2025; Borella, De Nardi, Torres Chain and Yang, 2026).

[4] See also Inkmann, Lopes and Michaelides (2011) and Hubener, Maurer and Rogalla (2014).

literature to research on how counterparty risk affects insurance demand.[5] A finance literature shows how financial frictions and capital regulation affect life insurers' pricing, capital, and exposure to market risk (Koijen and Yogo, 2015, 2016, 2022). We contribute to this literature by measuring subjective insurance nonpayment risk and documenting that it predicts actual insurance ownership.[6] Additionally, we embed this risky insurance into a structural model to show how nonpayment risk limits insurance ownership and affects life-cycle savings. Finally, we examine the consumer welfare gains from the existing, imperfect insurance available in the market, and contrast these gains to those achievable from risk-free insurance.

Impact of Social Insurance on Savings and Welfare. We also contribute to the study of how Social Security and Medicaid affect life-cycle savings and welfare. Previous research highlights how means-tested programs like Medicaid provide implicit insurance against medical expenses, rationalizing the low asset holdings of the poorest households (De Nardi, French and Jones, 2016). Braun, Kopecky and Koreshkova (2019) explore how public provision of insurance can crowd out demand for private insurance, and Conesa and Krueger (1999) study the effects of Social Security on savings and welfare over the life cycle. Recent work further explores how social insurance expansions alter household borrowing and financial resilience (Bornstein and Indarte, 2023) and affect inequality (Catherine, Lu and Paron, 2025a; Catherine, Miller and Sarin, 2025b). We add to this literature by examining how the public provision of insurance via Social Security and Medicaid interacts with the demand for a portfolio of imperfect private insurance products.

Consumer Beliefs and Economic Behavior. Finally, our work connects to the literature that measures consumer beliefs to understand economic behavior (Manski, 2004). Koşar and O'Dea (2023) provide an overview. In household finance, Giglio, Maggiori, Stroebel and Utkus (2021) link survey-elicited beliefs to administrative data, showing that beliefs about expected returns and rare-disaster probabilities predict equity portfolio allocations, and Bailey, Cao, Kuchler and Stroebel (2018) show that subjective beliefs about house prices affect housing tenure decisions. Our method of measurement builds on Luttmer and Samwick (2018), who study uncertainty surrounding Social Security benefits. We contribute to this literature by

[5] Doherty and Schlesinger (1990) show that nonperformance risk alters standard predictions of insurance demand, and several studies note that default probabilities can limit annuity market participation (Lopes and Michaelides, 2007; Pashchenko, 2013; Jang, Koo and Park, 2019). Cummins and Mahul (2003) and Mahul and Wright (2004) derive optimal coverage in static models in which the buyer's belief about total default differs from the insurer's and in which total default and partial payment are distinct risks. Bourgeon and Picard (2014) show that partial payment can arise in equilibrium, as insurers deter fraud by reducing payments on the claims they audit.

[6] Beshears, Choi, Laibson, Madrian and Zeldes (2014) document that survey respondents rank nonpayment risk as an important consideration limiting annuity purchases.

measuring beliefs about insurance nonpayment risk and demonstrating that these subjective expectations predict ownership of life-cycle insurance products. By embedding these elicited beliefs into a structural model, we quantify the welfare and saving implications of these consumer expectations.

2 Data

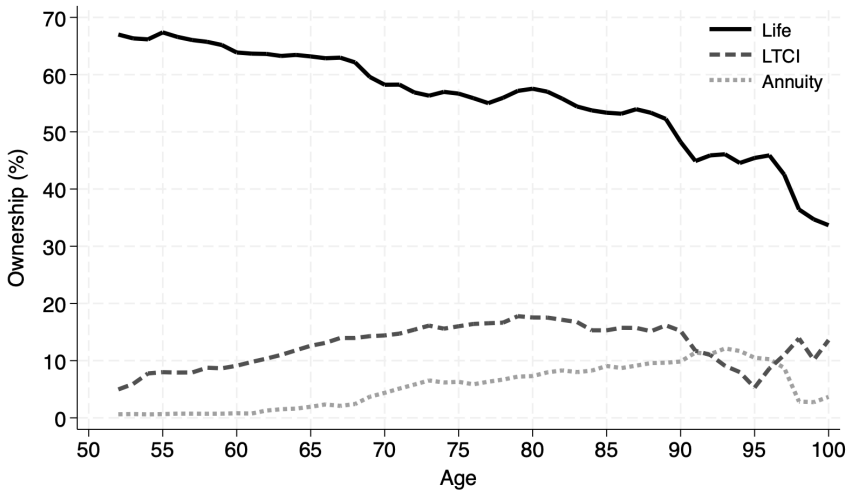
We use two sources of data. First, we use the Understanding America Study (UAS), a nationally representative internet panel of approximately 6,000 adults hosted by the University of Southern California. In May 2018 we surveyed 1,085 UAS respondents aged 45 and over about life-cycle insurance products. Second, we use the Health and Retirement Study (HRS), a nationally representative panel of more than 37,000 Americans older than age 50 with rich measures of health, income, assets, and insurance ownership from 1991 to the present.

We use the two sources in complementary ways. The UAS allows us to measure beliefs about insurance nonpayment risk and to correlate beliefs with insurance ownership and a rich set of individual characteristics. We also use the UAS to set initial values for the state variables of age, sex, income, health, wealth, and insurance ownership, in the structural model. In that sense, our structural model provides an analysis of the lives of our UAS survey respondents. The HRS is commonly used in the literature due to its high quality measurements and large sample size with a long panel dimension. Thus, we use the HRS to obtain empirical target moments to calibrate the structural model. After appropriate reweighting, both the UAS and HRS samples are representative of older Americans, so we utilize both sources of data without significant external-validity concerns.

2.1 Life-Cycle Patterns of Insurance Ownership

Figure 1 plots the cross-section of insurance participation rates by age from the 2014 HRS. Two patterns stand out. First, ownership of annuities and LTCI is rare: at age 65, roughly 12% of individuals own private LTCI and fewer than 5% own a private annuity, consistent with the classic annuity puzzle (Yaari (1965), Modigliani (1986)) and LTCI puzzle (Ameriks, Briggs, Caplin, Shapiro and Tonetti (2018); Braun et al. (2019)). Second, life insurance participation is much higher, around 60% at age 65, but declines steadily with age, consistent with earlier term policies purchased to insure children's welfare maturing over time (Hong and Ríos-Rull (2012)). Despite substantial health and mortality risks, Figure 1 indicates limited demand by older Americans for the annuity or LTC insurance products available in the market.

Figure 1: **Age Profile of Insurance Product Ownership Rates**



Note: Ownership rates for life insurance (solid black), long-term care insurance (dashed dark gray), and annuities (small-dash light gray) for US individuals. Source: Health and Retirement Study 2014.

2.2 The UAS Dataset

The UAS is an internet panel run by the USC Dornsife Center. Every two years, all participants complete a web-adapted HRS survey that provides harmonized data on health, work, finances, and behavioral traits. Linking our module to prior UAS surveys by respondent ID provides standard demographic and financial variables (age, sex, marital and labor-market status, wealth, and income).

Table 1 compares summary statistics of the UAS and corresponding HRS samples. Demographic and financial variables, including insurance ownership, are broadly similar across the two samples. UAS respondents, however, are slightly younger and less retired, with less wealth and higher incomes.

2.3 Insurance Survey Description

This section describes UAS survey 118 that we fielded to measure beliefs regarding insurance nonpayment risk.[7] We adapt the elicitation methodology developed by Luttmer and Samwick (2018) for Social Security benefits, to evaluate perceived risk in private insurance markets. Section 2.3.1 presents our two core measures. First, we measure the subjective distribution of partial nonpayment, defined as the fraction of the contracted payout received during a qualifying event. Second, we measure the subjective full-default probability, defined as

[7]The survey and collected data are available at `https://uasdata.usc.edu/survey/UAS+118`.

Table 1: **HRS and UAS Summary Statistics**

	HRS	**UAS**
Demographics		
Male	0.46	0.47
Age	66.9	61.1
Married	0.64	0.57
Education		
High school	0.64	0.68
Some college	0.23	0.18
College & above	0.13	0.14
Race		
White	0.83	0.81
Black	0.10	0.11
Hispanic & other	0.07	0.08
Health		
Good	0.74	0.79
Bad	0.23	0.17
LTC	0.03	0.04
Retired	0.46	0.34
Income ($000s)	69.7	100.6
Wealth ($000s)	582	461
Insurance ownership		
Annuity	0.04	0.05
Life insurance	0.61	0.58
LTCI	0.12	0.10
N	18,747	1,085

Note: We report sample means for each variable. Column (1) uses the HRS sample of individuals aged 50 and over; Column (2) uses respondents aged 45 and over from our UAS insurance survey. Health categories are self-reported. Insurance ownership variables are indicators for owning a private annuity, life insurance policy, or long-term care insurance policy. In both surveys, annuity ownership is measured as receiving annuity income. Observations are weighted using survey weights.

the likelihood that an insurer permanently ceases all future payments. Section 2.3.2 then presents certainty-equivalent valuations and auxiliary measures exploring the sources of these nonpayment beliefs.

2.3.1 Core Measures of Subjective Nonpayment Risk

Survey Design Overview. Our survey features three modules to elicit perceived nonpayment risk in annuity, life insurance, and LTC insurance products. Each product pays conditional on a qualifying event: survival for an annuity, death for life insurance, and needing help with at least two activities of daily living for LTC insurance. We randomize module order to mitigate and test for potential anchoring effects. Each module branches based on the respondent's current ownership status. If the respondent owns the product, we collect their

specific policy details (e.g., benefit size) and ask them to evaluate the nonpayment risk of their actual policy. For non-owners, we provide a standardized product description, randomly assign a hypothetical benefit size, and ask them to evaluate the best policy they believe they could currently purchase in the market. With a specific reference policy established, we elicit subjective nonpayment risk as follows. Appendix A reproduces the full wording of the survey questions.

Full-Default Probabilities. For each product $k \in \{\text{Ann}, \text{LI}, \text{LTC}\}$, we first ask respondents the annual probability of full default, defined as the event that the insurer permanently ceases all future payments through no fault of the policyholder. The prompt states that this may occur because the insurer fails, declares the contract void, or otherwise refuses to honor it, so the measure is a policy-level concept of full default rather than a measure of insurer insolvency alone. Respondents report this probability, denoted δ_i^k, on a 0-to-100 slider.

Annual Partial Nonpayment Distributions. We then elicit the perceived distribution of next year's payout conditional on a qualifying event. Following the bins-and-balls method of Delavande and Rohwedder (2008), respondents allocate 20 balls across three bins: no payment, a payment less than promised, and a payment at least as large as promised. Respondents then allocate the balls placed in the less-than-promised bin across five sub-bins of the payout expressed as a percentage of the contracted payment: 1–20%, 21–40%, 41–60%, 61–80%, and 81–99%. The exercise yields a respondent-specific subjective distribution Ω_i^k over the annual payout share $\omega^k \in [0, 1]$, from which we compute the mean $\mathbb{E}_i[\omega^k]$ and standard deviation $\sigma_i(\omega^k)$.

The three objects δ_i^k, $\mathbb{E}_i[\omega^k]$, and $\sigma_i(\omega^k)$ summarize the perceived probability of permanent contract failure and the uncertainty of payments while the contract remains in force. They are the belief measures used in the empirical analysis and in the structural model.

2.3.2 Auxiliary Measures of Subjective Nonpayment Risk

Next, we elicit certainty equivalents by asking respondents whether they would trade their risky insurance policy for a risk-free contract offering a guaranteed fraction of the promised payout. Respondents who perceive positive nonpayment risk make this choice exactly once for a randomly assigned guaranteed payout of 25%, 50%, 75%, 80%, or 90% of the promised amount. This single-shot elicitation means we do not observe certainty equivalents at the individual level, precluding the recovery of the full risk premia distribution. Instead, the aggregate acceptance rates across guaranteed payout thresholds map the population cumulative distribution function (CDF) of certainty equivalents. From this CDF, we compute

population-level summary statistics, including the mean certainty equivalent for each product and their implied risk premia.

A supplemental module, completed for one product after the core modules, elicits nonpayment beliefs conditional on a stock-market decline of 10% or 20% and the effort respondents expect to spend collecting a claim. A final module asks respondents to rank the three products from least to most risky and to rate their confidence in their answers. Respondents also identify the factors that influenced their beliefs (personal experiences, family/friends' experiences, government intervention, policy complexity, aggregate economic conditions, trust in salespeople, and other/free response). Appendix A reproduces these questions and Appendix B reports the results.

3 Perceived Nonpayment Risk

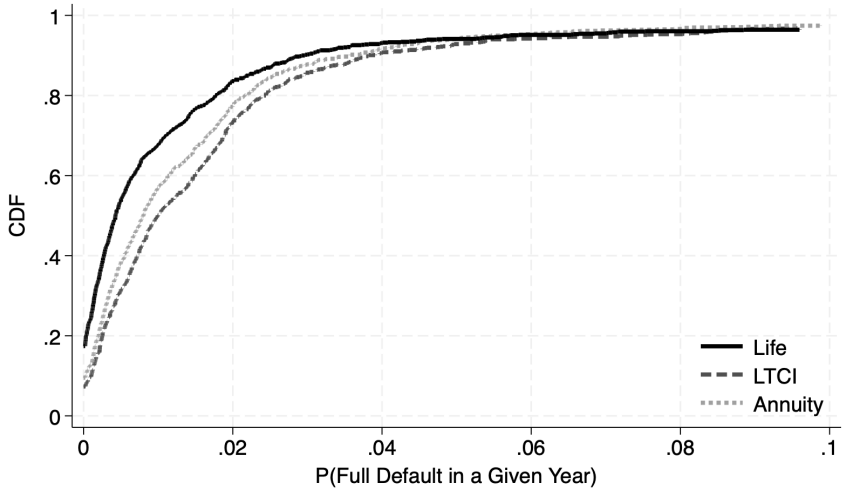
This section documents the elicited measures of perceived nonpayment risk, shows that these beliefs predict actual insurance ownership, and compares them with the available objective evidence on insurer impairment and claims-related nonpayment. All analyses apply UAS population weights to yield nationally representative estimates for Americans aged 45 and over.

3.1 Subjective Nonpayment Risk Measures

We document two primary metrics of perceived nonpayment risk: the annual probability of full default and the annual expected partial payout conditional on a qualifying event. Figure 2 presents the cross-sectional CDFs of subjective full-default probabilities. The data exhibit pervasive nonpayment concerns, with the vast majority of respondents assigning a positive probability to a full default. Only 19% of respondents report a zero full-default probability for life insurance, alongside 11% for annuities and 10% for LTCI. Responses reveal a clear ordinal ranking, with life insurance viewed as the least likely to default and LTCI as the most risky. At the median, the expected annual full-default probability is 0.3 percent for life insurance, 0.5 percent for annuities, and 1.0 percent for LTCI.

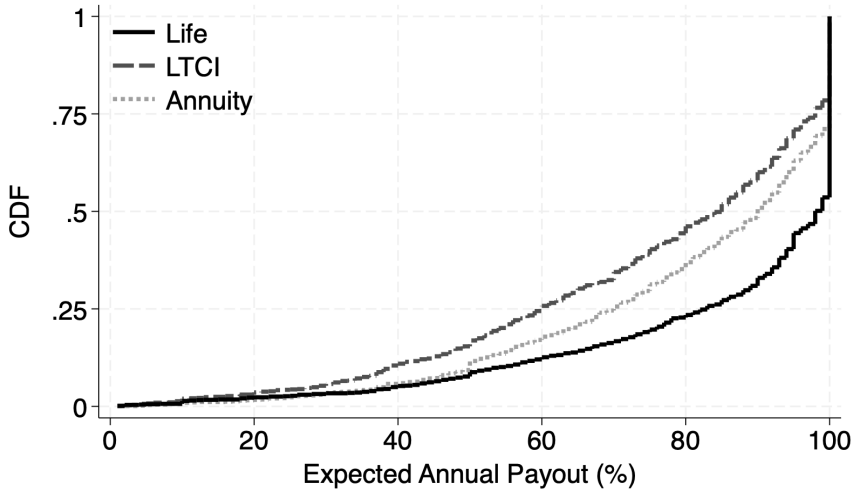
Figure 3 presents the cross-sectional CDFs for the mean of subjective annual payouts conditional on a qualifying event. Expected payouts are expressed as a fraction of the contracted payout. For life insurance, although around half of all respondents expect full payment, respondents on average expect to receive only 87% of the promised amount. Payout expectations are even lower for annuities and LTCI. For annuities, the mean expected payout

Figure 2: **Subjective Annual Probability of Full Default**



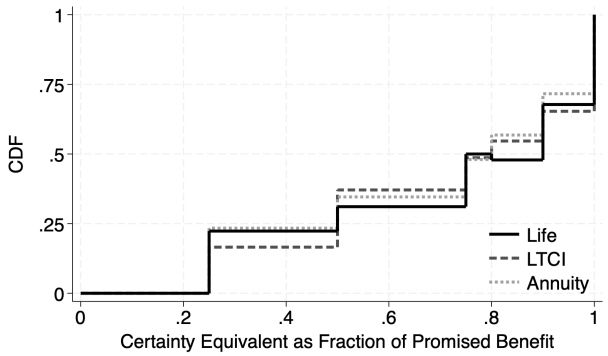
Note: Cross-sectional CDFs of subjective annual full-default probabilities, defined as the probability that an insurer permanently ceases all future payments.

Figure 3: **Subjective Annual Partial Payout Distribution**



Note: Cross-sectional CDFs of the mean of the subjective payout distribution conditional on a qualifying event. Payouts are expressed as a fraction of the contracted payout.

Figure 4: **Certainty Equivalent Distributions**



Note: CDFs of respondents' certainty equivalents, defined as the guaranteed fraction of the promised payout that renders a respondent indifferent to the risky policy. Each point on the CDF indicates the fraction of respondents who prefer that certain payout share to the risky policy.

is 82% and the median is 90%, while for LTCI the mean is 76% and the median is 85%.[8]

Together, these figures document our first key result: across all three products individuals perceive life-cycle insurance products as risky. Furthermore, we find life insurance is consistently viewed as the safest, LTCI as the riskiest, and annuities as intermediate in risk. Whatever the objective nonpayment probabilities, individuals act on these subjective beliefs.

3.2 Certainty Equivalents and Risk Premia

To evaluate how respondents value this uncertainty, we next examine the population distributions of certainty equivalents in Figure 4. Each point on these CDFs indicates the fraction of individuals who prefer a risk-free contract with that payout share to their current risky insurance policy. These measures are relatively similar across insurance types. At the median, respondents would trade a risky annuity, life insurance, or LTCI policy for a risk-free policy that guarantees 75–80% of the promised payout. This indicates that typical respondents are risk averse and willing to forgo a substantial share of the promised payout to eliminate nonpayment risk.

Table 2 reports the average expected payout and certainty equivalent for each insurance type. The difference between these measures is the implied risk premium. Risk premia are economically significant across all policies. The annuity risk premium, for instance, is nearly 8%, comparable to the well-documented equity risk premium. These results confirm that individuals view insurance as risky and require significant compensation to bear that risk.

[8]Appendix Figure B.1 reveals that respondents perceive substantial uncertainty around these payout expectations. On average, subjective payouts have a standard deviation of 17% of the promised amount for life insurance, 19% for annuities, and 20% for LTC insurance.

Table 2: **Insurance Product Risk Premia**

	Expected Value	Certainty Equivalent	Risk Premia
Life	87.16	81.43	5.72
Annuity	81.51	73.79	7.72
LTCI	76.17	72.90	3.27

Note: This table reports mean expected payout shares, certainty equivalent shares, and the implied risk premia for each insurance product. Payout values are expressed as percentage shares of the promised payouts. The risk premium equals the expected payout share minus the certainty equivalent share.

Appendix B additionally reports payout beliefs conditional on a stock-market decline. Respondents expect annuity and life insurance payouts to fall in a downturn, while their LTCI payout beliefs are unchanged. This pattern suggests that annuity and life insurance risk is more tied to insurer solvency and LTCI risk more to claim denials.

3.3 Subjective Nonpayment Risk and Insurance Ownership

The survey documents substantial perceived nonpayment risk. We next examine whether these subjective beliefs predict actual insurance ownership by regressing ownership indicators on our elicited risk metrics, controlling for demographic and other individual characteristics. Specifically, for each product $k \in \{\text{Ann, LI, LTC}\}$, we run the following probit regression across all respondents i:

$$\Pr\left(\text{Own}_i^k\right) = \Phi\Big(\alpha^k \;+\; \beta_1^k\, \mathbb{E}_i\left[\omega^k\right] \;+\; \beta_2^k\, \delta_i^k \;+\; \beta_3^k\, \sigma_i(\omega^k) \;+\; X_i'\gamma\Big) \tag{1}$$

where Own_i^k is an indicator for ownership of product k; $\Phi(\cdot)$ is the standard normal CDF; $\mathbb{E}_i\left[\omega^k\right]$ is the expected annual payout percentage share; $\sigma_i(\omega^k)$ is the standard deviation of the subjective payout distribution; δ_i^k is the annual full-default probability; and X_i is a vector of demographic and behavioral controls available in the UAS.[9] Table 3 reports the estimated average marginal effects, with p-values computed from robust standard errors.

The regression results indicate that perceived nonpayment risk is a deterrent to insurance ownership. For annuities, both a higher full-default probability and a larger standard deviation of perceived payouts significantly reduce the probability of ownership ($p < 0.001$). For life insurance, a higher expected payout predicts ownership ($p < 0.01$), and for LTC insurance,

[9]Demographic controls include age, wealth, income, education, marital status, race, health, household size, and retirement status, which have an intuitive relationship to ownership. Other controls include trust, cognitive score, financial literacy score, numeracy score, experienced fraud, risk aversion, propensity to plan, and early-life stock market returns. Conditional on the core risk metrics, these other covariates are generally insignificant. See Appendix B for an expanded table that reports estimated coefficients for all controls.

Table 3: **Regression of Insurance Ownership on Risky Insurance Beliefs**

	Own Annuity	**Own Life**	**Own LTCI**
Payout Expected Value ($\mathbb{E}_i[\omega^k]$)	-0.0002	0.0040**	0.0007
	(0.770)	(0.010)	(0.131)
Payout Standard Deviation ($\sigma_i(\omega^k)$)	-0.0025***	-0.0006	-0.0008
	(0.001)	(0.733)	(0.246)
Full-Default Probability (δ_i^k)	-0.0021***	-0.0012	-0.0022***
	(0.000)	(0.167)	(0.000)
N	1054	1045	1039
Pseudo R^2	0.267	0.214	0.175
Demographic Controls	Yes	Yes	Yes
Other Controls	Yes	Yes	Yes

p-values in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Note: Average marginal effects predicting product ownership via probit regression (1). Own Annuity equals one for respondents who own any annuity, including annuities that have not begun paying. Regressors include the subjective expected payout percentage share $\left(\mathbb{E}_i[\omega^k]\right)$, the payout distribution standard deviation $\left(\sigma_i(\omega^k)\right)$, and the annual full-default probability $\left(\delta_i^k\right)$. All three are in percent, so each coefficient is the change in the probability of ownership per percentage point. All specifications include demographic and supplementary individual controls (control coefficient estimates reported in Appendix Table B.1). p-values are reported in parentheses.

the full-default probability predicts lower ownership ($p < 0.001$). Although the remaining risk measures generally exhibit the expected signs, they are not statistically significant. Appendix B documents that these results are robust to excluding the extensive set of other controls. Those controls are almost always individually insignificant. Because beliefs were elicited after ownership decisions, these estimates do not establish causality, although a causal interpretation is plausible if nonpayment beliefs are persistent. Regardless, the robust correlation shows that the elicited beliefs capture economically meaningful variation rather than survey noise.

These estimates are also robust to the survey design and to response quality. Randomizing module order and hypothetical payout amounts leaves the core estimates unchanged, and 84% of respondents rank the three products in an order consistent with their quantitative responses, with results robust to restricting the sample to this group (Appendix B).

Table 4 quantifies the economic magnitude of these effects by comparing baseline predicted ownership to a counterfactual where insurance is perceived as risk-free ($\mathbb{E}_i\left[\omega^k\right] = 100$; $\sigma_i(\omega^k) = \delta_i^k = 0$). Eliminating perceived risk doubles participation rates for both annuities (from 12% to 24%) and LTC insurance (from 10% to 23%), and increases life insurance

Table 4: **Probit Counterfactual Ownership Rates with Safe Insurance**

	P(Own)	**P(Own | Safe Insurance)**
Annuity	.12	.24
Life	.57	.66
LTCI	.10	.23

Note: The first column reports predicted ownership probabilities using estimates from regression equation (1). The second column reports counterfactual results if all respondents perceive insurance products as risk-free, i.e., using the predicted values from regression equation (1) setting $\mathbb{E}[\omega^k] = 100$, $\sigma(\omega^k) = 0$, and $\delta^k = 0$.

participation from 57% to 66%. In sum, the survey data and probit estimates suggest that subjective nonpayment risk is substantial and meaningfully depresses insurance ownership.

However, these statistical counterfactuals implicitly hold all other individual characteristics constant. In practice, eliminating nonpayment risk would change behavior more broadly. For example, increased reliance on safe insurance might diminish the need for precautionary savings. This endogenous reduction in wealth would, in turn, feed back into insurance demand. To capture the joint determination of savings and insurance portfolios, and to evaluate the welfare implications of risky insurance, we develop a structural model in Section 4.

3.4 Subjective Beliefs and Objective Nonpayment Evidence

Nonpayment can occur if an insurer fails or if a solvent insurer voids the contract, denies or delays a claim, or pays less than contracted. The survey measures the buyer's belief about receiving the contracted benefit. Providing an empirical counterpart requires comprehensive evidence on both sources of nonpayment. No such measure exists, so the available evidence is partial.

Insurer failure is measured well and is rare. Across all AM Best-rated US insurers from 1977 to 2023, the average annual rate of financial impairment was 0.36 percent, falling to 0.11 percent or less among carriers rated A− or better.[10] Perceived annual full-default probabilities of 1.3 to 1.8 percent are several times this rate. Impairment, however, is not the full-default event our survey measures, for two reasons. First, impairment need not imply policyholder loss, since contracts may continue through rehabilitation or transfer and state guaranty associations cover obligations up to statutory limits. Those limits can bind for contracts of the size our survey describes, resolution can take years, and benefits are cut

[10]AM Best defines financial impairment as court-ordered conservation, rehabilitation, or liquidation, or a regulator's direction to delay or limit policy payments, and excludes impairments that occur after a rating is withdrawn. These rates are insurer-count weighted and combine life-and-health with property-and-casualty insurers (AM Best, 2024).

when assets run short (Appendix C). Second, full default on a policy also occurs when a solvent insurer declares the contract void or refuses to honor it. Impairment counts do not record these events. The gap can therefore reflect voiding and refusal by solvent insurers as well as beliefs about insurer failure that exceed its historical frequency. The data cannot separate the two.

Nonpayment by solvent insurers is much harder to measure. No dataset begins with the set of qualifying events and records whether the contracted benefit was received. Claim statistics condition on a filed claim and count denials for any reason, so they mix valid denials with wrongful ones. Regulatory examinations and settlements document specific payment failures but not their population frequency. Complaints to regulators and lawsuits are selected in both directions, since not everyone who complains or sues was owed payment and not everyone who was owed complains.

Among filed claims, denial is rare in life insurance and common in LTCI. Insurers report claim outcomes to state regulators in the Market Conduct Annual Statement, and the NAIC publishes state-level aggregates for 2021 to 2025 (National Association of Insurance Commissioners, 2026). Annuity filings contain no claim fields, so these statistics cover life insurance and LTCI. For life insurance, the median share across states of closed death claims denied, resisted, or compromised was 1.8 to 2.8 percent for non-cash-value policies and 0.6 to 0.7 percent for cash-value policies. For LTCI, the median denial ratio was 20 to 23 percent for initial eligibility requests and 6.5 to 8.1 percent for subsequent payment requests. The median share of eligibility determinations taking more than 60 days was 21.4 to 24.5 percent (Appendix Table C.1). These are gross rates and include justified denials. Clinical audits of LTCI denials adjudicated in 2007 and 2008 agreed with the insurer in 93 percent of cases (LifePlans, Inc., 2010). Denial counts also include requests that claimants do not pursue.

Claim statistics omit qualifying events for which no claim is filed, as when a policyholder or family gives up on the effort that collecting requires (Appendix A). There is clear evidence that some such benefits are wrongly withheld or paid only after review. Multistate examinations showed that life insurers had matched death records to stop annuity payments but not to pay death benefits. The resulting settlements led to $9.7 billion in payments to beneficiaries by 2018 (California Department of Insurance, 2018).

The available measures therefore quantify only one of the two sources of nonpayment, insurer failure. The magnitudes of nonpayment by solvent insurers and of benefits never collected remain unidentified.

Buyers face the same uncertainty and cannot resolve it from their own contracts. Life-insurance and LTCI claims arise years or decades after purchase, and an annuity's payments run for the rest of the holder's life. For all three products, the insurer therefore collects

premiums, or the annuity's full price, long before its performance on the tail can be observed. Buyers' beliefs instead draw on insurer reputation and on experience with insurance more broadly. Half of our respondents cite personal experience with insurers as a factor in their nonpayment beliefs, and over 40 percent cite the experience of family or friends, policy complexity, or trust in salespeople (Appendix Figure B.3). In other insurance markets, denials are common and many do not survive review. Of the disability claims the insurer Unum reassessed under a 2004 multistate settlement, 42 percent were reversed in whole or in part (Maine Bureau of Insurance, 2008). Of a sample of Medicare Advantage prior-authorization denials, 13 percent should have been approved under Medicare coverage rules (U.S. Department of Health and Human Services, Office of Inspector General, 2022). In homeowner claims from one multinational insurer across 28 countries, Gennaioli, La Porta, Lopez-de-Silanes and Shleifer (2022) find that 23 percent of claims are rejected outright and that claimants receive an average of 61 percent of the amount claimed.

Whether beliefs are accurate is a separate question from whether they are well measured, since subjective beliefs can differ from objective probabilities and still be measured well. The evidence for measurement quality is the invariance of the estimates to survey framing, the internal consistency of responses, and the predictive power of beliefs for ownership documented in the previous subsection and in Appendix B. Households choose portfolios based on their beliefs, so the model's positive predictions for insurance ownership and saving depend on beliefs rather than on objective probabilities. Its normative conclusions depend on both, because welfare depends on the payouts households actually receive. Since no comprehensive measure of objective nonpayment is available, we evaluate welfare under two bounding cases, one in which nonpayment risk is realized as perceived and one in which contracts always pay.

4 Model of Life-cycle Saving and Insurance Demand

This section develops a structural model of life-cycle insurance demand to quantify how nonpayment risk and institutional frictions shape saving, portfolio choice, and consumer welfare. We embed multiple insurance products into an otherwise standard incomplete-markets life-cycle model of consumption and saving with health and mortality risk and borrowing constraints (cf. De Nardi et al. (2010), Lockwood (2018), Ameriks et al. (2020)). The model incorporates realistic insurance market features, including subjective nonpayment risk, pricing loads, illiquidity (sale discounts), and quantity restrictions such as age cutoffs and short-sale limits. We discipline nonpayment beliefs using our UAS insurance survey data, while remaining product characteristics are calibrated to statistics of market contracts taken from the literature. Standard life-cycle dynamic variables—including income, wealth, and

health and mortality risks—are calibrated to match HRS data. We first outline the model environment and calibration, before presenting the quantitative results in Section 5.

Model Overview. We develop a model of individual choice with exogenous prices that are calibrated to match real-world prices. Time is discrete and each period is a year. Individuals are heterogeneous in exogenous state variables: age (t), health status (s_t), sex (f), and an income-age profile (y, with element y_t). Health and mortality are the main nonfinancial risks in the model. An individual's health status can be good, poor, in need of long-term care, or dead. It evolves according to a Markov transition matrix that is a function of age, sex, health status, and a measure of permanent income. Health directly affects flow utility via a nonhomothetic health-state-dependent utility function, out-of-pocket medical expenditures, and insurance prices. Individuals enter each period holding risk-free bonds (b_t) and a portfolio of insurance contracts (x_t^k) for $k \in \{\text{LI}, \text{LTC}, \text{Ann}\}$. The annual premia for life insurance ($p_{i,t}^{\text{LI}}$) and LTCI ($p_{i,t}^{\text{LTC}}$) are locked in at the time of purchase and depend on contemporaneous individual states. At the beginning of the period health and insurance nonpayment shocks are realized. Given their updated states and shocks, individuals choose how much to consume, save, and invest across a portfolio of insurance products or to use means-tested government services. Individuals do not equalize marginal utility across dates and states due to insurance nonpayment risk, price loads, quantity limits, and borrowing constraints.

4.1 Model Details

Demographics and Health. The model is populated by heterogeneous individuals indexed by age $t \in \{45, 46, \ldots, 99\}$ and sex $f \in \{0, 1\}$, with $f = 1$ denoting female. Each period, an individual's health status is given by $s_t \in \{0, 1, 2, 3\}$, corresponding respectively to normal health, poor health, requiring long-term care, and death. An individual requires LTC if they need and receive help with activities of daily living (ADLs).[11] Health evolves according to a Markov transition matrix $\pi(s_{t+1} \mid s_t, t, f, y)$ which is a function of current health, age, sex, and permanent income. We allow for transition probabilities to differ for those in the bottom three permanent-income quintiles versus the top two, capturing the well-documented wealth gradient in health and longevity. Additionally, there are out-of-pocket medical costs, $\xi \sim \Xi(\xi \mid s_t, t, f)$. This lognormally distributed cost is a negative shock to liquid wealth, with a mean that increases in age and worse health states.

[11] An individual requires LTC if they receive help in at least two of five ADL categories (bathing, eating, dressing, getting in/out of bed, and walking), as measured by the adl5h variable in the HRS.

Preferences. Households have time-separable, health-state-dependent nonhomothetic preferences for consumption, c_t, and a warm-glow bequest motive. Flow utility u_s is

$$u_s(c_t) = \theta_{s,t}^{-\gamma} \frac{(c_t + \kappa_s)^{1-\gamma}}{1-\gamma}$$

where γ is the coefficient of relative risk aversion, κ_s controls the degree to which expenditure in health state s is a luxury or a necessity, and $\theta_{s,t}$ scales marginal utility. Following Ameriks et al. (2020), this specification allows for nonhomothetic utility in the LTC state ($\kappa_2 \neq 0$), analogous to the standard nonhomothetic treatment of bequests ($\kappa_3 \neq 0$) developed by De Nardi (2004). Because this framework nests the more common model ($\kappa_2 = 0$), the data can inform the importance of nonhomothetic LTC utility, placing long-term care risk on equal footing with the bequest motive to explain measured life-cycle saving patterns.

We depart from the prior literature by allowing the bequest scale $\theta_{3,t}$ to vary with age. This parsimoniously captures age-dependent bequest motives, reflecting, for instance, the changing economic needs of children over the life cycle. For ages $t \geq 80$, the bequest parameter is constant at $\theta_{3,t} = \bar{\theta}_3$. For ages $45 \leq t < 80$ it evolves according to:

$$\theta_{3,t}^{-1} = \theta_{3,45}^{-1} + \left(\bar{\theta}_3^{-1} - \theta_{3,45}^{-1}\right) \left(\frac{\exp\Big(\rho\,(t-45)/(80-45)\Big) - 1}{\exp(\rho) - 1} \right). \tag{2}$$

The parameter $\theta_{3,45}^{-1}$ anchors the initial strength of the bequest motive, while the parameter ρ controls the curvature of its life-cycle profile.[12]

Income. Surviving individuals receive an exogenous, deterministic annual income, y_t. This income comprises post-tax labor earnings, Social Security benefits, and defined-benefit pension payouts. Using HRS data, we estimate five life-cycle income profiles via quintile regressions and assign individuals to a profile based on their measured income. We abstract from idiosyncratic income shocks primarily to reduce the computational burden of the dynamic portfolio choice problem. Because our analysis focuses on older cohorts whose income depends more on annuitized retirement payouts like Social Security and pensions, muting income risk provides a tractable approximation of their late-in-life economic environment.

Assets. The core innovation of our structural model is the joint saving and insurance portfolio decision. Individuals allocate wealth across four assets. The first is a standard

[12] As ρ tends to 0 the bequest motive becomes linear in age. Our calibration finds $\theta_{3,45}^{-1} > \bar{\theta}_3^{-1}$ and $\rho > 0$, producing a bequest motive that is decreasing and strictly concave in age for ages $45 \leq t < 80$, thereby concentrating the decline at older ages.

risk-free bond, $b_t \geq 0$, which pays a constant return r and is subject to a no-borrowing constraint. For tractability, given the complexity of the insurance asset space, we use this single liquid-wealth variable to encompass all assets other than insurance.

Second, individuals can purchase three types of insurance products, indexed by $k \in \{\text{LI}, \text{LTC}, \text{Ann}\}$: immediate life annuities, whole life insurance, and long-term care insurance. Each year, individuals choose their desired start-of-next-period holdings for each insurance product, denoted by $\tilde{x}^k_{t+1}$.

The insurance products are distinguished by the health states s in which they pay out. Let χ^k be a 4×1 vector of dividends for asset k, where row s specifies the unit payout in state s: $\chi^{\text{Ann}} = [1, 1, 1, 0]'$; $\chi^{\text{LI}} = [0, 0, 0, 1]'$; $\chi^{\text{LTC}} = [0, 0, 1, 0]'$. Thus, annuities pay while alive, life insurance pays upon death, and LTCI pays when in need of LTC. These payout vectors, combined with the stochastic health dynamics, determine the covariance of these insurance asset payouts.

Individuals take exogenous prices as given. Insurance prices are a function of age, health, and sex and are calibrated to match real-world prices in Section 4.3.

Annuities are purchased via a lump-sum payment at unit price $p^{\text{Ann}}_{t,s,f}$. In contrast, life insurance and LTCI require ongoing annual premia. These premia are locked in at initial purchase based on an individual's contemporaneous age, health, and sex at prices $p^{\text{LI}}_{t,s,f}$ and $p^{\text{LTC}}_{t,s,f}$, making the contracted premia endogenous state variables ($p^{\text{LI}}_{i,t}$ and $p^{\text{LTC}}_{i,t}$). To maintain computational tractability, individuals may hold at most one life insurance and one LTCI policy at a time. To adjust these holdings, they must let their existing contract lapse and purchase a new policy at current prices.

The economic environment features several institutional frictions. Short-selling is prohibited ($\tilde{x}^k_{t+1} \geq 0$), and new policies cannot be purchased after an exogenous maximum age T, reflecting insurer concerns over adverse selection. Additionally, annuities can be sold for cash value, but at a discounted price $p^{\text{Ann}}_{t,s,f}\left(1 - \lambda^{\text{Ann}}_{-}\right)$.

Insurance Nonpayment Risk. Consistent with our survey measurements, insurance contracts are subject to two forms of nonpayment risk. First, an insurer may fully default on the contract with probability δ^k. Absent default, next period's insurance holdings equal the chosen quantities. Under a full default, holdings fall to zero without any sale proceeds:

$$x^k_{t+1} = \begin{cases} \tilde{x}^k_{t+1} & \text{with probability } 1 - \delta^k \\ 0 & \text{with probability } \delta^k. \end{cases} \tag{3}$$

For life and LTC insurance, a full default destroys the locked-in premium advantage. To restore the lost coverage, the individual must repurchase the policy at prevailing spot prices. Those prices are higher because the individual is older. For annuities, future payouts are lost with zero recovery of the initial lump-sum purchase amount.

Second, insurers may partially default on promised payouts. In any given period, the realized payout is only a fraction $\omega_t^k \in [0,1]$ of the promised payout, where $\omega_t^k \sim \Omega^k(\cdot)$.

To preserve computational tractability, we abstract from belief heterogeneity. All individuals share a common subjective probability distribution over these nonpayment shocks, matched to the central tendencies of our survey data.

Government Insurance. The government provides a state-dependent consumption floor, $\bar{c}_s$. To approximate government means-testing rules, we constrain participating individuals to enter the subsequent period with zero assets, $b_{t+1} = x_{t+1}^k = 0$. For individuals in normal or poor health, the consumption floor is $\bar{c}_0 = \bar{c}_1 = \bar{c}$. For those who need LTC ($s = 2$), the government provides $\bar{c}_2$. This parameter captures the value individuals place on government-provided LTC services, such as Medicaid-funded nursing homes.

Intertemporal Budget Constraint. Collecting these components, we write the intertemporal budget constraint as:

$$\begin{aligned}\frac{1}{(1+r)}b_{t+1} = b_t + y_t + v_{s_t}\sum_k \omega_t^k x_t^k \chi^k - c_t - \xi_t - p_{i,t+1}^{\text{LTC}}\tilde{x}_{t+1}^{\text{LTC}} - p_{i,t+1}^{\text{LI}}\tilde{x}_{t+1}^{\text{LI}} \\ - (\tilde{x}_{t+1}^{\text{Ann}} - x_t^{\text{Ann}})p_{t,s,f}^{\text{Ann}}\left(1 - \lambda_-^{\text{Ann}}\mathbb{I}_{(\tilde{x}_{t+1}^{\text{Ann}} < x_t^{\text{Ann}})}\right).\end{aligned} \tag{4}$$

The right-hand side of the constraint balances available resources against period expenditures. Resources consist of initial bond holdings b_t, income y_t, and total realized insurance payouts. We define v_{s_t} to be a 1×4 indicator vector that identifies the individual's current health state, so $v_{s_t}\sum_k \omega_t^k x_t^k \chi^k$ captures total insurance payouts net of any partial nonpayment shocks. These resources fund consumption c_t, out-of-pocket medical expenses ξ_t, and the annual premia for life and LTC insurance policies. The prices for the annual premia are locked in at the purchase price as long as the individual is not choosing to change the quantity owned ($\tilde{x}_{t+1} = x_t$). The final term captures annuity transactions. Adjustments are made at the prevailing spot price $p_{t,s,f}^{\text{Ann}}$, subject to the illiquidity discount λ_-^{Ann} when selling.

Full Decision Problem. The value function for an individual with states $S_t := \{s_t, y, f, b_t, x_t^{\text{Ann}}, x_t^{\text{LI}}, x_t^{\text{LTC}}, p_{i,t}^{\text{LTC}}, p_{i,t}^{\text{LI}}\}$ is $V_t(S_t) = \max\{V_t^{\text{Gov}}(S_t)\ ,\ V_t^{\text{Mkt}}(S_t)\}$. The value of using means-tested government services is:

$$V_t^{\text{Gov}}(S_t) = u_s(\bar{c}_s) + \beta \mathbb{E}\left[V_{t+1}(s_{t+1}, y, f, 0, 0, 0, 0, 0, 0)\right].$$

Alternatively, individuals remaining in the market choose consumption, savings, and desired insurance portfolio (c_t, b_{t+1}, $\tilde{x}_{t+1}^{\text{Ann}}$, $\tilde{x}_{t+1}^{\text{LI}}$, and $\tilde{x}_{t+1}^{\text{LTC}}$) to maximize their value:

$$\begin{aligned} V_t^{\text{Mkt}}(S_t) = \max\, & u_s(c_t) + \beta \mathbb{E}\left[V_{t+1}(S_{t+1})\right] \\ \text{s.t.} \quad & \frac{1}{(1+r)} b_{t+1} = b_t + y_t + v_{s_t} \sum_k \omega_t^k x_t^k \chi^k - c_t - \xi_t - p_{i,t+1}^{\text{LTC}} \tilde{x}_{t+1}^{\text{LTC}} - p_{i,t+1}^{\text{LI}} \tilde{x}_{t+1}^{\text{LI}} \\ & \qquad - (\tilde{x}_{t+1}^{\text{Ann}} - x_t^{\text{Ann}}) p_{t,s,f}^{\text{Ann}} \left(1 - \lambda_-^{\text{Ann}} \mathbb{I}_{(\tilde{x}_{t+1}^{\text{Ann}} < x_t^{\text{Ann}})}\right). \end{aligned}$$

Full default forces realized next-period insurance holdings to zero:

$$x_{t+1}^k = \begin{cases} \tilde{x}_{t+1}^k & \text{with probability } 1 - \delta^k \\ 0 & \text{with probability } \delta^k. \end{cases}$$

The locked-in annual premia for life and LTCI reset to the spot price if the contract changes quantity, either voluntarily or through default:

$$p_{i,t+1}^k = \begin{cases} p_{i,t}^k & \text{if} \quad \tilde{x}_{t+1}^k = x_t^k \\ p_{t,s,f}^k & \text{if} \quad \tilde{x}_{t+1}^k \neq x_t^k \end{cases} \quad \text{for} \quad k \in \{\text{LI}, \text{LTC}\}.$$

There are insurance no-short-sale and maximum-issue-age constraints, $\tilde{x}_{t+1}^k \geq 0$ and $\tilde{x}_{t+1}^k \leq x_t^k$ for $t > T$. Health, medical costs, and nonpayment risk are distributed according to $\pi(s_{t+1} \mid s_t, t, f, y)$, $\Xi(\xi \mid s_t, t, f)$, and $\Omega^k(\omega^k)$. Prices $p_{t,s,f}^k$ and r are taken as given.

Numerical Solution. Because our incomplete-markets framework precludes the analytical solutions available in complete-market settings (e.g., Koijen et al. (2016)), we solve the individual choice problem numerically via backward induction using a grid search over a discretized state space. To simulate the model, we draw initial conditions directly from the empirical joint distribution of the UAS survey. Specifically, we assign each simulated individual their observed age, sex, income, wealth, and initial insurance holdings. We then simulate the model forward, allowing these individuals to dynamically optimize their consumption, savings, and insurance portfolios over the remainder of their life. Model statistics are computed using UAS population weights. Appendix E.1 details the solution and simulation procedures.

4.2 Discussion of Portfolio Choice and Incomplete Markets

Incomplete-market features. Households can provide for longevity, long-term care, and bequests through liquid saving or insurance. Liquid wealth can finance any of them, while insurance pays only in particular states. Under complete markets (e.g., Koijen et al. (2016)), individuals could perfectly hedge these concerns using annuities, life insurance, and LTCI, eliminating the need for precautionary bond holdings. In contrast, our incomplete-market framework introduces several risks, costs, and constraints that may depress insurance demand and push individuals toward self-insurance via personal savings.

Insurance nonpayment risk (δ^k and ω^k) is the most critical friction and the main source of market incompleteness. Insurance pays in the states where marginal utility is highest, so a partial or total shortfall in the payment reduces resources in exactly those states. No market exists to hedge it. Anticipating this uninsurable shortfall deters insurance purchases, even when perfect insurance for that state would be highly valued. Full default also destroys the value built up in the contract: the remaining payments bought with an annuity's up-front price and, for life and LTC insurance, the premia locked in at an earlier age.

The second major friction is insurance pricing. In practice, prices are not actuarially fair. They embed substantial loads above discounted state probabilities. Furthermore, annuity liquidations face steep sale discounts. A related wedge arises because the return on household wealth, r, differs from the insurer pricing rate, r_I. We interpret r as the effective return on the broader household portfolio represented by bonds in the model, including housing and higher-yielding financial assets. When $r > r_I$, purchasing insurance requires households to forgo a higher return on liquid wealth. This wedge is distinct from the explicit product loads and grows with the time between paying for insurance and receiving benefits. That time is long for all three products, since annuities require a large front-loaded purchase and life and LTC insurance premia may be paid long before benefits are received. Absent nonpayment risk, the insurance products span all health states. Nonetheless, price loads and the return wedge would prevent the equalization of marginal utility across states.

Third, uninsurable out-of-pocket medical expenses generate a precautionary saving motive (see, e.g., De Nardi et al. (2010)). Individuals may be reluctant to lock their wealth into insurance contracts that cannot be easily liquidated to cover immediate medical cost shocks. Finally, we impose a maximum age at which individuals are able to purchase new policies. In practice, insurers routinely reject older applicants, possibly due to adverse selection concerns.[13] These features preserve a role for liquid wealth even though the model contains

[13]For example, Hendren (2013) reports that LTCI applications for individuals over age 80 are automatically rejected, while MetLife notes that 98% of annuity purchases occur before age 85, and most life insurance companies stop issuing new policies around age 80–90.

products designed to insure each major life-cycle risk.

Wealth and permanent income. Wealth is a central determinant of the choice between market insurance and self-insurance. As liquid wealth rises, households become better able to finance adverse states directly and avoid exposure to nonpayment risk, price loads, and contract illiquidity. Existing wealth can also finance bequests, reducing the marginal value of life insurance. The relationship need not, however, be monotone. This is especially clear for LTCI because the model includes a means-tested government consumption floor. Low-wealth households may expect to rely on public care, middle-wealth households may value LTCI most because they wish to avoid public care but cannot comfortably self-insure, and sufficiently wealthy households can increasingly finance care out of pocket. For annuities, greater wealth creates more resources that could be annuitized but also expands the capacity to self-insure longevity risk.

Permanent income adds a second channel. It raises resources, like wealth, but it also shifts health and survival probabilities. Insurers in the model do not price these differences. Conditional on priced characteristics, higher-income households tend to live longer and therefore receive relatively favorable annuity prices, since they expect to collect payments for more years than the pooled price implies. The same selection works in the opposite direction for life insurance. The effect on LTCI is less clear. Greater longevity creates more opportunity to enter the LTC state, while better health transitions may reduce the probability of requiring care at a particular age.

State-dependent preferences and bequest motives. Insurance demand depends not only on the probability that a contract pays, but also on the marginal utility of resources in that state. The health-state-dependent utility specification allows the value of consumption to differ across normal health, poor health, and the LTC state, while nonhomotheticity permits consumption to behave more like a necessity in some states than in others. An insurance product becomes more attractive when it transfers resources into a state in which marginal utility is high. The LTC utility parameters therefore govern whether households strongly value private expenditure while requiring care or are relatively willing to rely on the government floor.

The age-dependent bequest motive is particularly important for life insurance. When the desire to leave resources is stronger at younger ages, an early death creates a large shortfall between desired and available bequests, raising the value of life insurance. Death is less likely at these ages, but this also makes coverage cheaper, while locked-in premia further strengthen the incentive to purchase when young. As the bequest motive weakens

with age and households accumulate wealth, life-insurance demand declines. The effect on LTCI is ambiguous. LTC benefits can preserve other assets for heirs because entry into LTC is associated with elevated subsequent mortality. At the same time, a stronger bequest motive raises life-insurance purchases, which directly provide resources in the death state and may reduce the need to use LTCI for this purpose. Which force dominates depends on the calibrated parameter values.

Joint portfolio choice. The value of each of the four assets depends on holdings of the others. The insurance products can be substitutes or complements. Life insurance and LTCI can both support bequests, and so may substitute for each other. Life insurance can make annuitization more attractive by separately protecting bequests. LTCI can do the same by reducing the need to hold liquid wealth for care expenses. Portfolio choice therefore depends on the covariance of each product's payout with marginal utility given the rest of the portfolio, the cost of transferring resources across states and dates, and the option value of retaining liquid wealth.

4.3 Model Calibration

We first calibrate some parameters of the model externally and then use the simulated method of moments (SMM) to calibrate the remaining parameters.

Standard Parameters External Calibration. We use HRS data to measure income, wealth, health, and medical costs. We estimate five income-age profiles using sex-specific quintile regressions. This income measure includes post-tax labor earnings, Social Security benefits, and defined-benefit pension payouts. Bonds, b_t, encompass all non-insurance assets, including financial and housing wealth.[14] We set the government consumption floor for individuals not requiring LTC to \$5,000 ($\bar{c} = 5$), consistent with standard values in the literature (e.g., De Nardi et al. (2016)). See Appendix D for details on constructing y_t, b_t, $\pi(s_{t+1} \mid s_t, t, f, y)$, and $\Xi(\xi \mid s_t, t, f)$.

Insurance Parameters External Calibration. We next calibrate the parameters that determine insurance product characteristics, reported in Table 5. To avoid sensitivity to outliers, we truncate the UAS insurance survey responses at the bottom and top 2%. We then use the mean of these trimmed responses to calibrate the full-default probabilities

[14]Financial wealth is the sum of liquid assets (checking, saving, money market, and certificate of deposit accounts) and investment assets (brokerage, mutual fund, education-saving, individual retirement, and defined-contribution retirement accounts).

δ^k. For the partial nonpayment distributions, $\Omega^k(.)$, we discretize the outcome space into four equally likely events and set the associated partial-payment values to match the mean and standard deviation of the empirical subjective annual payout distributions. This yields $\omega_t^{\text{Ann}} \in \{.50, .78, .94, 1\}$, $\omega_t^{\text{LI}} \in \{.58, .92, 1, 1\}$, and $\omega_t^{\text{LTC}} \in \{.48, .71, .88, 1\}$.[15]

In calibrating prices, we confront individuals with the prices they would face in the real-world market, remaining agnostic about their supply-side determinants. In practice, prices are not actuarially fair. They embed substantial loads above discounted state probabilities. Brown and Finkelstein (2011) and Hong and Ríos-Rull (2012) report real-world market prices in terms of loads above actuarially fair prices. The actuarially fair price is computed to equal the discounted expected lifetime payouts of the contract, conditional on the purchasing individual's age, health, sex, and the corresponding health transition probabilities. Therefore, we define a product-specific load, λ^k, such that multiplying the actuarially fair price by $1 + \lambda^k$ yields the empirical market price. We compute these actuarially fair prices using a risk-free interest rate of $r_I = 2.5\%$, in line with Brown and Finkelstein (2007, 2011), and Koijen and Yogo (2015).

To match observed market prices, we apply the loads reported by Brown and Finkelstein (2011) and Hong and Ríos-Rull (2012): 20 percent for annuities ($\lambda^{\text{Ann}} = 0.20$), 25 percent for life insurance ($\lambda^{\text{LI}} = 0.25$), and 32 percent for long-term care insurance ($\lambda^{\text{LTC}} = 0.32$). For annuity liquidations, industry reports (Christian, 2025) suggest sale discounts of 20–40 percent, so we set the sale penalty to 35 percent ($\lambda_{-}^{\text{Ann}} = 0.35$). We interpret λ^k as reduced-form wedges that summarize supply-side forces, including administrative and distribution costs, capital charges, adverse selection, expected lapsation revenues, and markups. This approach generates an empirically accurate choice environment without explicitly modeling insurer operations.

Remaining Parameters SMM Calibration. Having fixed the exogenous state processes and insurance product characteristics, we jointly calibrate the remaining preference and return parameters using the simulated method of moments (SMM). These parameters include γ, θ_2, κ_2, $\bar{c}_2$, $\theta_{3,45}$, ρ, $\bar{\theta}_3$, κ_3, β, and r. We allow the rate of return on individual savings to differ from the risk-free rate used to price insurance, as the single saving asset in the model proxies for the broader, higher-yielding portfolio of assets individuals hold in the data.

The SMM procedure targets three sets of empirical moments: (i) the 25th, 50th, and 75th percentiles of the wealth distribution by 5-year age bins for ages 50–84; (ii) the average ownership rate of annuities (4%), life insurance (61%), and long-term care insurance (12%);

[15] Outcomes are set to the 12.5, 37.5, 62.5, and 87.5 percentiles of the empirical subjective annual payout distributions. This procedure closely matches the empirical means and standard deviations.

Table 5: **Baseline Calibration - Insurance Products**

	Annuities	**Life**	**LTCI**
Full Default (δ^k)	.014	.013	.018
Annual Payout Fraction ($E[\omega]$)	.815	.871	.761
Annual Payout Uncertainty ($\sigma(\omega)$)	.194	.174	.195
Insurance Loads (λ^k)	.2	.25	.32
Annuity Sale Discount (λ_{-}^{Ann})	.35	–	–
Max Purchase age (T)	80	80	80

Note: Payout beliefs calibrated to match surveyed subjective nonpayment belief distributions. Insurance loads from Brown and Finkelstein (2011) and Hong and Ríos-Rull (2012). Maximum purchase age from Hendren (2013). Annuity sale discount from industry report Christian (2025).

and (iii) mean responses to the nine strategic survey questions (SSQs) from Ameriks et al. (2020). These SSQs place individuals in hypothetical scenarios and ask them to make choices that are designed to identify preferences. We target the extensive margin of insurance ownership because survey data reliably captures binary participation status. Self-reported face values and benefit levels are likely measured with greater error. The SMM minimizes the distance between these empirical moments and their simulated counterparts (see Appendix E for calibration details).

To manage the computational burden of solving the life-cycle portfolio problem over a high-dimensional state space and parameter space, we employ a multi-stage optimization routine. We initialize the utility parameters (γ, θ_2, κ_2, $\bar{c}_2$, $\bar{\theta}_3$, and κ_3) at the estimated values from Ameriks et al. (2020) and conduct a grid search over the remaining parameters (β, $\theta_{3,45}$, ρ, r). Then, we hold β, $\theta_{3,45}$, ρ, and r at their best-fit values, and search over the initially-fixed preference parameters to improve the model fit. Because this second stage yielded substantial improvements only via the risk aversion parameter (γ), our final optimization stage jointly calibrates $(\gamma, \beta, \theta_{3,45}, \rho, r)$, holding the remaining parameters fixed at their initial Ameriks et al. (2020) values.

Calibrated Parameter Values. Table 6 reports the calibrated parameter values. A value of $\gamma = 1.5$ implies a moderate degree of risk aversion, which limits LTCI ownership by moderating the desire to insure the high-marginal-utility LTC state. The bequest motive for younger individuals, controlled by $\theta_{3,45}^{-\gamma}$, scales the marginal utility of a bequest at age 45 by approximately 180 relative to death at age 80. This generates a powerful incentive for younger individuals to hold life insurance to protect against the rare event of early death. The curvature parameter $\rho = 1.5$ sustains this strong bequest motive through middle age, causing it to substantially taper only after age 70.

Table 6: **Baseline Calibration - Preferences and Returns**

LTC luxury ($000s)	$\kappa_2 = -37.44$	**Time preference**	$\beta = 0.9575$
LTC marginal	$\theta_2 = 0.67$	**Risk aversion**	$\gamma = 1.5$
Bequest luxury ($000s)	$\kappa_3 = 7.83$	**Govt LTC floor** ($000s)	$\bar{c}_2 = 77.43$
Old bequest marginal	$\bar{\theta}_3 = 1.09$	**Govt floor** ($000s)	$\bar{c} = 5$
Young bequest marginal	$\theta_{3,45} = 0.034$	**Household bond return**	$r = 0.05$
Bequest life-cycle shape	$\rho = 1.5$	**Insurer discount rate**	$r_I = 0.025$

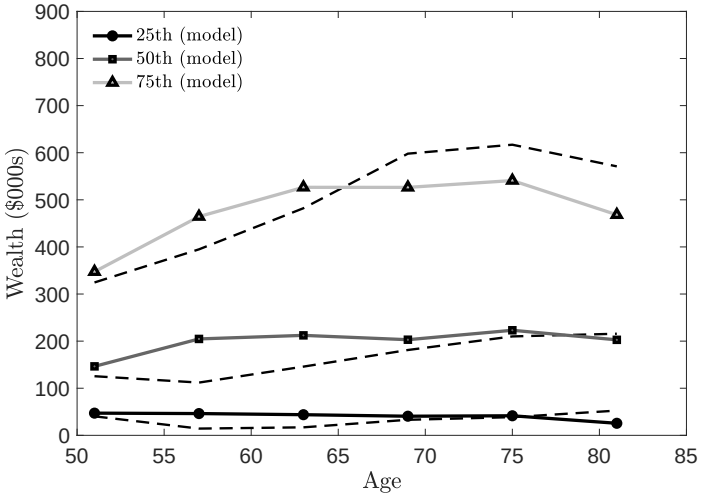
Note: Baseline calibration for parameters governing preferences and returns. Marginal utility in state s scales with $\theta_s^{-\gamma}$, so a smaller θ means a higher marginal utility of resources in that state. See text for calibration procedure.

Time preference also plays a critical role in shaping insurance demand. The calibrated $\beta = 0.9575$ and $r = 0.05$ put $\beta(1 + r)$ just above one, which keeps wealth from falling late in life as the HRS wealth profiles require (Appendix E). A higher β also shifts weight away from the early-death state, where the bequest motive is strongest and life insurance is most valuable, toward longevity risk, LTC risk, and consumption later in life. Households self-insure these later risks with accumulated wealth, avoiding the loads, nonpayment risk, and illiquidity of the insurance products. In the calibrated model, a higher β therefore lowers life insurance demand and, more weakly, LTCI demand. Finally, the household return $r = 0.05$ exceeds the insurer pricing rate $r_I = 0.025$, so a dollar moved into insurance forgoes 2.5 percentage points a year. This wedge tilts portfolios toward self-insurance, most for annuities, whose front-loaded purchase locks in the lower rate for longest. Appendix E.4 provides further details on the sensitivity of model outcomes to these parameter values.

5 Model Results: Insurance Ownership and Welfare

This section presents results from our calibrated structural model. We begin in Section 5.1 by establishing that the calibrated model accurately matches both targeted moments and untargeted patterns of insurance ownership. The remainder of this section details four core findings. First, Section 5.2 quantifies how subjective nonpayment risk and price loads significantly depress insurance ownership and increase wealth due to precautionary saving. Second, Section 5.3 evaluates the consumer welfare gain provided by real-world insurance products relative to a no-insurance benchmark. At baseline, the market yields only modest welfare gains equivalent to 0.6% of lifetime consumption. However, a market offering insurance with zero nonpayment risk and zero loads generates substantial consumer welfare gains of 4.1%, reconciling our baseline findings with complete-market benchmarks. Third, Section 5.4 decomposes these gains, revealing two key insights: LTCI generates the largest welfare

Figure 5: **Wealth Moments: Model and Data**



Note: Wealth levels by age for the 25th, 50th, and 75th percentiles of the distribution. Data moments (dashed lines) are from the HRS; model moments (solid lines) are from the baseline calibration.

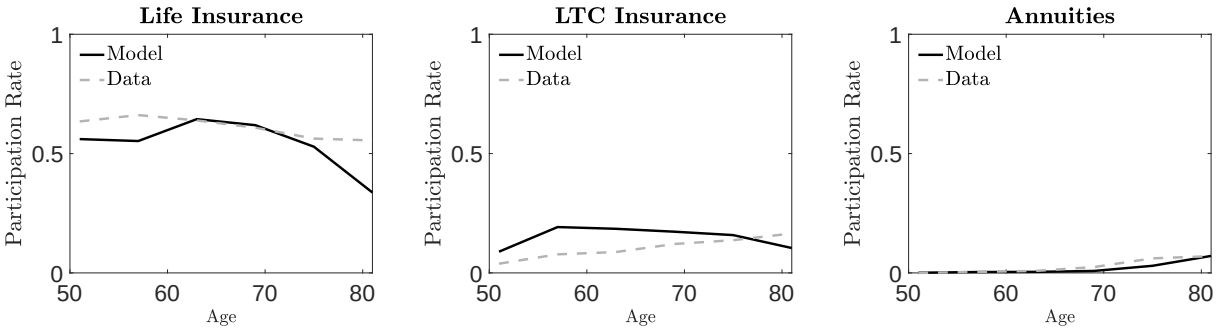
improvement despite lower ownership rates than life insurance, and partial nonpayment risk is more than three times as detrimental to welfare as full-default risk. Finally, Sections 5.5 and 5.6 quantify the impact of government-provided LTCI and annuities, via Medicaid and Social Security, on private insurance ownership and welfare.

5.1 Model Fit

Figure 5 shows that the model replicates both the average level and the life-cycle profile of wealth across the wealth distribution. Although the calibration only targeted average insurance participation rates, the model successfully generates several untargeted features of insurance demand. Figure 6 demonstrates our fit to the life-cycle pattern of insurance ownership.[16] Furthermore, Table 7 shows that the model captures portfolio composition in terms of both the number of insurance products owned and the correlations in ownership across products. Nearly 90% of individuals either hold no private insurance at all or only a single life insurance product. When they do hold a second product, it is most often LTC insurance, consistent with the positive correlation between Life and LTCI ownership.

[16]Model-implied insurance holdings can differ from initial holdings for several reasons. Households may rebalance their portfolios immediately in the initial model period, subject to the relevant transaction costs. In addition, holdings at a given age are averaged across all simulated agents who reach that age, including those initialized at younger ages and simulated forward.

Figure 6: **Insurance Ownership: Model and Data**



Note: Insurance participation rates by age. Dashed lines are empirical rates from the HRS; solid lines are from the baseline calibrated model.

Table 7: **Nontargeted Insurance Ownership Moments**

A. *Number of Products Owned*	**0**	**1**	**2**	**3**
Model	41%	46%	12%	1%
Data	35%	55%	9%	1%
B. *Ownership Correlations*	**(LI, LTCI)**	**(LI, Ann)**		**(LTCI, Ann)**
Model	0.15	-0.03		0.10
Data	0.08	-0.02		0.06

Note: Panel A reports the fraction of individuals owning zero to three insurance products. Panel B reports the pairwise correlation of joint participation between two insurance products. Data are from the HRS; model results are from the baseline simulation.

Table 8: **Insurance Ownership and Savings: Impact of Risky Insurance and Loads**

	Data	Baseline	Safe Insurance	Zero Loads	Zero Loads & Safe Insurance
A. *Ownership Rate*					
LI	61%	55%	73%	71%	84%
LTCI	12%	13%	43%	29%	55%
Ann	4%	2%	8%	3%	10%
B. Δ *Personal Savings*					
25th percentile	–	−36%	−48%	−42%	−55%
50th percentile	–	−16%	−27%	−21%	−33%
75th percentile	–	−4%	−16%	−10%	−22%

Note: Panel A reports aggregate ownership rates for three insurance products under alternative scenarios. Panel B reports percentage changes in wealth across the distribution (see Figure 5), relative to values from a no-insurance benchmark, averaged across age with population weights. Baseline is the calibrated model; Safe Insurance sets default probabilities to zero ($\omega^k = 1$, $\delta^k = 0$); Zero Loads sets product-level loads λ^k to zero; Zero Loads & Safe Insurance combines both adjustments.

5.2 Determinants of Insurance Market Participation and Savings

We now quantify how product market frictions shape individuals' saving and insurance portfolio allocations. Table 8 reports outcomes under four scenarios: the *Baseline* calibrated specification; *Safe Insurance*, which sets all subjective nonpayment probabilities to zero; *Zero Loads*, which eliminates the load markups ($\lambda^k = 0$); and a combined scenario *Zero Loads & Safe Insurance*.[17]

Panel A of Table 8 reports aggregate ownership rates for annuities, life insurance, and LTCI. Our baseline calibrated model matches the empirical aggregate participation rates well. The next three columns quantify how real-world product characteristics depress insurance demand. For life insurance, eliminating nonpayment risk significantly increases participation, with ownership rising to 73%. The 25% load also severely suppresses demand: setting loads to zero raises ownership to 71%. Together, zero loads and zero nonpayment risk increase life insurance participation to 84%. LTCI is even more affected by these frictions: eliminating nonpayment risk more than triples ownership (from 13% to 43%), removing loads doubles it (to 29%), and removing both expands participation to 55%. In contrast, the effects on annuities are muted: ownership rises to 8% without nonpayment risk, 3% with zero loads, and 10% when both frictions are removed.

Panel B of Table 8 quantifies the impact on savings by reporting the percentage change in

[17]We evaluate the zero-load counterfactual without taking a stance on whether these empirical loads generate pure profits or cover necessary operating costs. Rather, this adjustment allows us to contrast our findings more closely to complete market benchmarks, where actuarially fair prices fundamentally alter the consumer choice problem.

wealth at the 25th, 50th, and 75th percentiles relative to an economy with no private insurance. In the baseline scenario, access to insurance reduces wealth most dramatically for poorer individuals (a 36% decline at the 25th percentile), who in the no-insurance counterfactual rely heavily on precautionary savings to hedge against health and mortality risks. In contrast, wealthier individuals, who are more likely to have adequate savings across states, show smaller reductions (16% at the median and 4% at the 75th percentile).

The decline in wealth is more pronounced as frictions are removed. The largest drop occurs when both loads and nonpayment risk are eliminated. Two mechanisms drive this result. First, as cheaper and safer insurance products become more attractive investments, individuals substitute out of risk-free bonds and into formal insurance. Second, improved access to state-contingent protection weakens the precautionary motive to hold liquid wealth. Consequently, when these product market frictions are removed, individuals purchase more insurance and draw down their wealth much more aggressively in old age.

Overall, both subjective nonpayment risk and high loads significantly depress insurance ownership rates and generate substantial increases in precautionary savings.

5.3 Welfare Gains from Life-Cycle Insurance Products

In this section we quantify the welfare gains provided by insurance products with varying characteristics. We report the consumption-equivalent welfare gain relative to a no-insurance benchmark, aggregated across individuals with UAS survey weights.[18]

Up to now, we have not needed to take a stand on whether individuals' subjective beliefs are rational. An individual's portfolio allocation rule is a function of their state variables and subjective beliefs, regardless of whether those beliefs reflect the true objective probability distribution. Furthermore, because purchases typically occur long before qualifying events, payout realizations do not meaningfully alter the state variables at the time of purchase. Consequently, insurance ownership is virtually identical in a world of rational expectations and in a world in which payouts are always made. For welfare, however, the distinction matters, because receiving payouts is obviously better than not receiving them. As documented in Section 3.4, no data provide a comprehensive measure of policy-level or claim-level nonpayment risk. We therefore evaluate welfare under two bounding cases. First, we simulate the model under rational expectations, where nonpayment risk realizes according to subjective

[18]Consumption-equivalent welfare is the percent decrease in expenditure in every health state in the considered scenario that makes an individual indifferent between the no-insurance benchmark and the considered scenario (Appendix E.3). In the no-insurance benchmark, households relinquish their existing life and long-term care insurance policies and cease paying the associated premia, while existing annuity holdings are converted into a lump-sum payment equal to the present discounted value, computed at the insurer discount rate, of their expected future payouts.

Table 9: **Welfare Gains From Insurance**

	Baseline	**Safe Insurance**	**Zero Loads**	**Zero Loads & Safe Insurance**
A. *Rational Beliefs*				
Δ Welfare	0.6%	2.3%	1.8%	4.1%
B. *Non-Rational Beliefs*				
Δ Welfare	1.7%	2.3%	3.4%	4.1%

Note: Consumption-equivalent welfare gains relative to a no-insurance benchmark. Panel A assumes rational expectations; Panel B assumes zero objective nonpayment risk. Subjective beliefs are the same across panels within each scenario. Baseline is the calibrated model; Safe Insurance sets perceived and objective default probabilities to zero; Zero Loads sets product-level loads λ^k to zero; Zero Loads & Safe Insurance combines both adjustments.

nonpayment probabilities. Second, we simulate the model with zero actual nonpayment risk. In both cases nonpayment risk beliefs are identical, set to their baseline measured values.

Panel A of Table 9 shows that, when expectations are rational, baseline welfare gains from the existence of these insurance products are modest at 0.6% of lifetime consumption. This reflects the fact that nearly half of individuals optimally choose not to own any formal insurance (Table 7), relying instead on self-insurance and public insurance. Furthermore, those who own insurance are left with products that sometimes do not pay when promised, exactly when individuals have revealed they highly value payouts by their purchase of insurance in the first place. Removing frictions produces substantial welfare improvements. Eliminating nonpayment risk almost quadruples the baseline gains to 2.3%, removing loads nearly triples them to 1.8%, and eliminating both simultaneously yields a 4.1% welfare gain.

This pattern underscores the central result of the model: large welfare gains arise when insurance products are available at actuarially fair prices and are free of nonpayment risk because individuals can effectively hedge substantial health risks they value insuring. In contrast, insurance products with high loads and nontrivial risk of nonpayment discourage individuals from purchasing these products at a substantial loss of consumer welfare.

Panel B of Table 9 reports welfare gains when actual nonpayment risk is zero. Because portfolio allocation rules depend solely on subjective beliefs, portfolio choices are identical to the rational expectations case, but ex-post welfare is evaluated using simulations in which promised payouts are always made. This generates a mechanical welfare increase: baseline gains rise from 0.6% under rational expectations to 1.7% simply because individuals experience zero defaults. More interestingly, comparing this 1.7% gain with the 2.3% gain from eliminating both subjective and objective nonpayment risk highlights the role of subjective beliefs. In both scenarios, the objective product characteristics are identical and payouts

are guaranteed. However, when individuals accurately perceive that there is no nonpayment risk, they purchase significantly more insurance (Table 8 Panel A). This shift toward using the available products at scale generates an additional 0.6% welfare gain. Here, the welfare gained simply by correcting beliefs is as large as the rational-expectations baseline value of the products existing in the first place.

Across both perspectives, the core conclusion remains clear. Removing product market frictions—whether eliminating loads or removing actual or perceived nonpayment risks—yields substantial welfare gains by allowing individuals to efficiently insure major life-cycle health and mortality risks. Equivalently, the welfare costs of not owning real-world insurance products are much lower than the costs of not owning idealized products that complete markets.

5.4 The Sources of Welfare Gains

To determine how each type of insurance contributes to consumer welfare improvements, Table 10 reports the gains from introducing each product individually, relative to the zero-insurance benchmark.[19] In the baseline economy, the majority of the 0.6% total welfare gain stems from LTCI, even though life insurance is held by roughly four times as many individuals. Isolating life insurance yields only a 0.1% welfare gain, whereas isolating LTCI generates a 0.5% gain. Annuities alone have a negligible contribution. This hierarchy persists when nonpayment risk is eliminated, since LTCI suffers from the highest perceived default risk. However, when price loads are removed, life insurance accounts for a larger share of the total welfare gain. Since life insurance is already perceived as the safest product and has the highest baseline ownership, its welfare contribution share is particularly sensitive to price reductions.

Nonpayment risk takes two forms: annual partial nonpayment risk and full-default risk. Table 10 decomposes the welfare cost associated with each type of risk. When all insurance products are available, the annual partial nonpayment risk accounts for 76% of the total gains from removing all nonpayment risk. Eliminating only the annual partial nonpayment risk delivers a 1.3% gain over the baseline compared to the 1.7% gain from removing all nonpayment risk. Turning to individual products, life and LTC insurance are paid for gradually by annual premia, so if a policy defaults the individual simply ceases future premia payments. As a result, the primary cost of a full default is the loss of the preferential premium price that was locked-in at initial purchase. In contrast, a full default on an annuity is substantially more costly because the individual loses the entire upfront lump-sum purchase

[19] For simplicity of exposition, the remainder of the paper focuses on the rational expectations case. The main takeaways are robust to the case with zero objective nonpayment risk (holding subjective beliefs constant at measured levels); see Appendix F for these results.

Table 10: **Welfare Gains from Individual Insurance Products**

	Baseline	No Annual Nonpayment Risk	Safe Insurance	Zero Loads	Zero Loads & Safe Insurance
LI	0.1%	0.6%	0.7%	1.0%	1.7%
LTCI	0.5%	1.1%	1.3%	0.7%	1.5%
Ann	0.0%	0.1%	0.2%	0.2%	0.7%
All	0.6%	1.9%	2.3%	1.8%	4.1%

Note: Consumption-equivalent welfare gains relative to a no-insurance benchmark (rational beliefs case). Each row isolates the introduction of only the specified product. The final row introduces all three products and reproduces Panel A from Table 9. Baseline is the calibrated model; No Annual Nonpayment Risk sets the annual partial nonpayment risk to zero ($\omega^k = 1$), but leaves full-default probabilities δ^k at baseline levels; Safe Insurance sets all default probabilities to zero; Zero Loads sets product-level loads λ^k to zero; Zero Loads & Safe Insurance sets all nonpayment probabilities and loads to zero.

payment. As a result, full-default risk accounts for roughly half of the total nonpayment risk costs for annuities.

In summary, LTCI generates 0.5% of the total 0.6% baseline welfare gain provided by all insurance products. Additionally, removing annual partial nonpayment risk has more than three times the welfare impact of eliminating full-default risk. This is especially true for life and LTC insurance, which are the products that deliver the vast majority of welfare gains across all scenarios.

5.5 The Effect of Medicaid on Ownership and Welfare

Our baseline calibration reflects the empirically low participation rate in the private LTCI market. Because high loads and nonpayment risks depress the value of private insurance, individuals rely on two substitute channels to insure against LTC shocks. The first is Medicaid, which provides a guaranteed consumption floor, $\bar{c}_2$. Due to its fixed care quality and strict means testing, this public safety net disproportionately benefits poorer individuals and crowds them out of the private LTCI market. The second is self-insurance. Wealthier individuals, who are less likely to rely on Medicaid, instead accumulate substantial precautionary savings.[20]

Given these factors, we examine the extent to which public LTC generosity crowds out private insurance demand. Panel A of Table 11 evaluates private LTCI participation as we progressively reduce the public consumption floor ($\bar{c}_2$).[21] As Medicaid LTC support declines,

[20]In the model, 27% of individuals in the bottom wealth quartile use Medicaid at some point in their lives, compared to just 8% in the top quartile. Heavy reliance on Medicaid extends into the second wealth quartile, where 26% of individuals eventually use public LTC. This occurs because the consumption floor of $\bar{c}_2 = 77.43$ is high relative to their typical wealth holdings, which would otherwise be exhausted by roughly two years of spending when in need of LTC.

[21]We omit improvements to the government budget from lower public LTC expenditures.

Table 11: **The Effects of Government LTC Provision (Medicaid)**

	Baseline $\bar{c}_2 = 77.43$	$\bar{c}_2 = 60$	$\bar{c}_2 = 40$
A. *Ownership Rate*			
LI	55%	53%	49%
LTCI	13%	29%	54%
Ann	2%	3%	3%
B. *LTCI Ownership Rate*			
0–25th Wealth Percentile	8%	12%	42%
25–50th Wealth Percentile	25%	42%	72%
50–75th Wealth Percentile	13%	29%	56%
75–100th Wealth Percentile	10%	35%	46%
C. *Welfare Effects*			
Aggregate	-	-1.3%	-4.0%
0–25th Wealth Percentile	-	-1.4%	-5.7%
25–50th Wealth Percentile	-	-1.7%	-6.1%
50–75th Wealth Percentile	-	-1.3%	-2.7%
75–100th Wealth Percentile	-	-0.7%	-1.2%

Note: Columns correspond to alternative levels of the public long-term care consumption guarantee ($\bar{c}_2$). Panel A reports model-implied insurance participation rates. Panel B reports private long-term care insurance ownership rates across the wealth distribution. Panel C reports consumption-equivalent welfare changes relative to the Baseline, in aggregate and by wealth percentile, using population weights.

there is a strong substitution into private LTCI. For example, reducing $\bar{c}_2$ to 40 more than quadruples the LTCI ownership rate from 13% to 54%. As Panel B highlights, this effect is most pronounced in the second wealth quartile, where participation rises from 25% to 72%. These middle-wealth individuals lack the substantial precautionary savings necessary to fully self-insure, yet they possess enough resources to afford private insurance and avoid diminished public care.

Panel C reports the corresponding welfare effects relative to the baseline economy. Lowering $\bar{c}_2$ has large aggregate costs, approximately 4% of lifetime consumption when $\bar{c}_2 = 40$. These costs are heavily concentrated among poorer individuals. For the bottom half of the wealth distribution, welfare falls by roughly 6%, whereas wealthier individuals are partially insulated because they rely more on personal savings than public care.

In summary, while the baseline provision of public LTC substantially crowds out private LTCI participation, it generates significant consumer welfare, particularly for individuals in the lower half of the wealth distribution.

Table 12: **The Effects of Social Security**

	Data	Baseline	No Social Security	No Annuity Frictions & No Social Security
A. *Ownership Rate*				
LI	61%	55%	19%	30%
LTCI	12%	13%	14%	20%
Ann	4%	2%	4%	19%
B. *Aggregate Bequests*				
Δ vs. Baseline	-	-	23%	5%
C. *Welfare Gains*				
Δ Welfare	-	-	2.0%	4.1%

Note: Panel A reports insurance ownership rates in the data and across model scenarios. Panel B reports the percent change in aggregate bequests (both intentional and incidental) relative to Baseline. Panel C reports consumption-equivalent welfare changes relative to Baseline. No Social Security removes Social Security payments and rebates their present value to individuals as a lump-sum transfer. No Annuity Frictions sets annuity loads and default probabilities to zero.

5.6 The Effect of Social Security on Ownership and Welfare

Social Security benefits are embedded in our baseline estimates of life-cycle income profiles. Because Social Security functions essentially as a publicly provided annuity, we now evaluate how its existence affects private insurance ownership, wealth accumulation, and welfare. To do so, we simulate a counterfactual where we remove Social Security payments from the baseline estimation of income and instead credit individuals with the present value of the payments they would have otherwise received.[22]

While one might expect individuals to replace the lost public annuity with private annuities, Panel A of Table 12 shows that they do not. Private annuity participation remains low at 4%. This occurs because market frictions make private annuities an unattractive asset. When we eliminate both loads and nonpayment risk for annuities, ownership rises to 19%.

Relative to the baseline, the primary portfolio shift under the lump-sum counterfactual is a reduction in life insurance. Instead of purchasing formal life insurance, individuals self-insure using their increased liquid wealth. Panel B highlights this wealth accumulation: aggregate bequests rise by 23%, reflecting increases in both intentional and incidental bequests. Because individuals hold more wealth later in life to self-insure against longevity risk, early mortality generates larger bequests. However, when annuity market frictions are removed this effect is

[22]We discount these payments using the risk-free rate r_I, the same rate used to price private insurance products, so the lump sum equals the market value of the forgone benefits. Catherine et al. (2025b) use this risk-free valuation as their baseline measure of Social Security wealth and note that liquidity-constrained households may value the benefits below it. See Appendix D.2 for details.

muted, with bequests rising by only 5%.

Panel C of Table 12 reports the welfare implications of the Social Security counterfactual. When we replace Social Security with an upfront lump-sum transfer, aggregate welfare improves by 2.0% over baseline. Households cannot borrow against future Social Security benefits, so receiving them up front relaxes the borrowing constraint and lets households smooth consumption intertemporally. Welfare gains rise to 4.1% when private annuity market frictions are simultaneously removed. This 2.1% marginal improvement is significantly larger than the 0.7% gain from removing annuity frictions in the baseline model (Table 10). Intuitively, without Social Security crowding out demand, individuals benefit more from improved private annuities, reflected in both higher participation rates and welfare.

Overall, replacing Social Security with an up-front wealth transfer substantially reduces life insurance participation and has minimal effects on LTCI and private annuity participation. Welfare is higher on average in this counterfactual, driven by the relaxation of borrowing constraints, even though individuals lose the de facto Social Security annuity and its private market alternatives carry perceived nonpayment risk and loads.

6 Conclusion

Older Americans face substantial health, mortality, and longevity risks, yet participation in private insurance markets—particularly for annuities and long-term care insurance—remains puzzlingly low. While previous literature demonstrates that individuals would highly value perfect, complete-market insurance, we show that this lack of participation is largely driven by the imperfect nature of real-world products. A primary contribution of this paper is demonstrating that subjective nonpayment risk is a pervasive concern among consumers and a major deterrent to insurance ownership. Using a purpose-designed survey, we document that, on average, individuals expect to receive 76%–87% of promised payouts conditional on a qualifying event, and perceive annual full-default probabilities ranging from 1.3% to 1.8% across products. Validating these measures, we find that elicited beliefs predict actual insurance ownership.

Embedding these beliefs into a structural life-cycle model reveals that nonpayment risk and high pricing loads severely depress insurance demand, encourage precautionary saving, and reduce consumer welfare. Baseline access to existing, imperfect insurance products generates a modest welfare gain of 0.6% of lifetime consumption and reduces median wealth by 16%. Eliminating nonpayment risk and pricing loads substantially increases insurance participation rates—from 55% to 84% for life insurance, 13% to 55% for LTCI, and 2% to 10% for annuities. Consequently, the welfare gain from these idealized insurance products

rises to 4.1% and the reduction in median wealth to 33%. Finally, we document how public insurance programs like Medicaid and Social Security affect private insurance demand and welfare.

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A UAS Insurance Survey Questions

The complete survey and collected data are available at `https://uasdata.usc.edu/survey/UAS+118`. This appendix reproduces the principal questions from the survey, using the annuity module as an illustrative example. The life insurance and LTC insurance modules follow the same structure with product-specific qualifying events and benefits.

Fielding and Response. Our survey was fielded in May 2018. Among invited panelists, the response rate was 82%.[23] In total, 1,104 started and 1,085 completed the survey. Average completion time was 17 minutes. Item non-response was minimal: 1,045 answered all life insurance items, 1,054 all annuity items, and 1,039 all LTCI items. To maximize sample size, we do not restrict analyses to a common respondent set across products.

A.1 Core Measures of Subjective Nonpayment Risk

Full-Default Probabilities. We first ask respondents about the probability of full default. Specifically, we ask the following:

> Suppose that you own an annuity that promises to pay $[20,000] each year for the rest of your life. Suppose further that you never trade this annuity for cash and hold the contract until the end of your life.
>
> We are now interested in the percent chance that in a given year the annuity becomes worthless due to no fault of your own. This means that the annuity permanently stops making payments. This might occur if the insurance company goes out of business, they claim you violated a clause in the contract, or they ruled the policy void for some other reason.
>
> What is the percent chance this occurs?

Respondents input their perceived probability using a 0-to-100 slider, accompanied by dynamic text that explains the meaning of their current selection.

Annual Partial Nonpayment Distributions. We then elicit the perceived distribution of payouts conditional on a valid qualifying event. Following the bins-and-balls method of

[23] This is a lower bound, because the link closed when the purchased quota was met.

Delavande and Rohwedder (2008), respondents use a visual interface to allocate balls that represent probability mass across bins that define potential outcomes. On the initial screen, respondents distribute 20 balls across three bins, for no payment, a partial payment, or the full promised amount.

> Suppose that you own an annuity that promises to pay $[20,000] each year for the rest of your life. We would now like to focus on what might happen just during the next calendar year.
>
> You have been given 20 balls to put in the following bins. Each bin describes a scenario that involves the annuity payment that you are supposed to receive next year. The more likely you think a bin is, the more balls you should put in that bin.
>
> What do you think will happen to the annuity payment next year?
>
> I will receive no payment/I will receive a payment less than I am supposed to receive/I will receive a payment at least as large as I am supposed to receive

Respondents then reallocate any probability mass assigned to the partial-payment event across five sub-bins (ranging from 1–20% to 81–99%). These bins represent the realized payout as a percentage of the contracted amount.

> You put [13] ball(s) in the bin marked "I will receive a payment less than I am supposed to receive." Please distribute those balls in the following bins. The more likely you think a bin is, the more balls you should put in that bin.
>
> If you do receive a payment that is less than you are supposed to receive, how much do you think you would get?

This procedure elicits the full subjective payout distribution, allowing us to compute the mean and standard deviation of perceived annual payouts as a fraction of the contracted amount.[24]

[24]Moment calculations assume all probability mass is located at the midpoint of each bin.

A.2 Auxiliary Measures of Subjective Nonpayment Risk

Certainty Equivalent. The respondent's expected payout is filled in from their previous responses. The guaranteed value is randomized.

The way you put balls into various bins shows that you expect to receive about [83]% of your annuity payment next year. It also shows that you could receive more or less than [83]% of the promised payment.

Let's call this distribution of possible payments, as described by you using the bins and balls, your "uncertain payments." So, your uncertain payments are whatever payments you think you might receive next year.

We are now interested in how you value having a contract with no uncertainty. Imagine a contract that is guaranteed to pay [75]% of your annuity payment with no risk of the insurance company not paying out as promised. This is like having all 20 balls on this certain percentage. This contract is unbreakable and cannot be changed by anybody. This contract has no risk, but is guaranteed to pay less than the full promised amount of your original contract.

Would you rather have:

- Guaranteed payment equal to [75]% of the annuity payment you are supposed to receive.

  or

- Uncertain payments around an expectation of [83]% of the annuity payment you are supposed to receive?

Beliefs Conditional on Aggregate Financial Conditions. Each respondent completes this supplement for a single insurance product after completing the core modules. To prevent framing effects from contaminating the primary data, the supplement corresponds to the final product in the respondent's randomized sequence. Respondents are prompted with a stock market decline of a randomly assigned magnitude and then, using an elicitation structure identical to the core modules, report both the full-default probability and the subjective partial payout distribution conditional on this shock.

> We are now interested in the chance that the insurance company will meet the policy's obligations under different economic conditions.
>
> Specifically, suppose that the stock market decreases by [10/20]% next year.

Effort in Collecting Payment.

> In general, how much effort do you think you will need to put in to receive the annuity payments you are promised?
>
> For example, this could include you or your family members doing paperwork, talking with claims officers, talking with doctors, hiring lawyers, or other such activities.
>
> Please choose one of the following:
>
> - No effort at all.
> - A small amount of effort.
> - A medium amount of effort.
> - A large amount of effort.

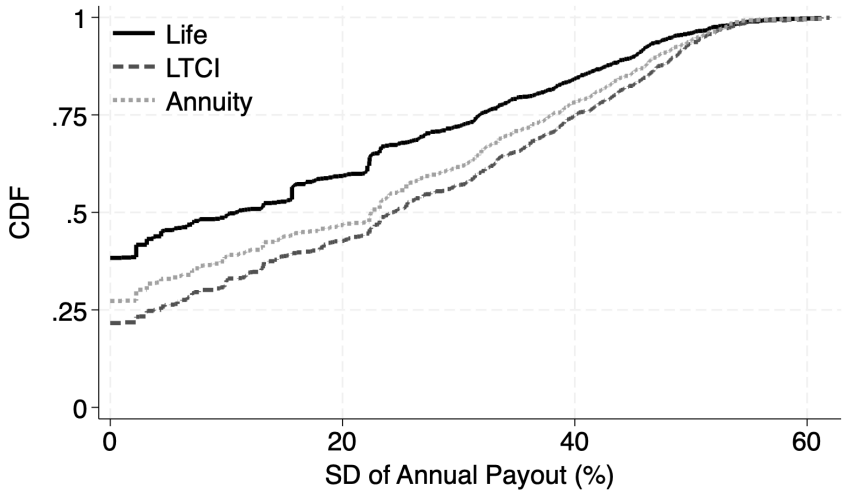
Final Module. The survey concludes with a module asking respondents to reflect on their answers. To test for internal consistency, we ask respondents to qualitatively rank life, annuity, and LTC insurance from least to most risky. We define a set of responses as internally inconsistent if none of the core quantitative risk metrics is largest for the insurance type qualitatively ranked as the most risky. Next, respondents identify the factors

that influenced their beliefs: personal experiences, family/friends' experiences, government intervention, policy complexity, the aggregate economic conditions, trust in salespeople, other/free response. Finally, we elicit respondents' confidence in their answers and the amount of prior thought they had given to both insurance purchases and nonpayment risk.

Conversion of Lifetime Full-Default Probabilities to Annual Rates. In the survey, we randomly assigned respondents to one of two possible framings for the elicitation of the full-default probability: "We are now interested in the percent chance that" (i) "in a given year the annuity becomes worthless due to no fault of your own," and (ii) "the annuity becomes worthless due to no fault of your own at any point before the end of your life." We express all results in annualized terms, which requires converting responses from frame (ii) into annual rates. To do so, first we use the respondent's age, sex, permanent income, and health status, together with the health-state Markov transition matrix $\pi(s_{t+1} \mid s_t, t, f, y)$, to compute their life expectancy. Second, we divide the full-default lifetime probability by their expected years of remaining life.

B Perceived Nonpayment Risk: Additional Results

Figure B.1: **Standard Deviation of Subjective Annual Payouts**

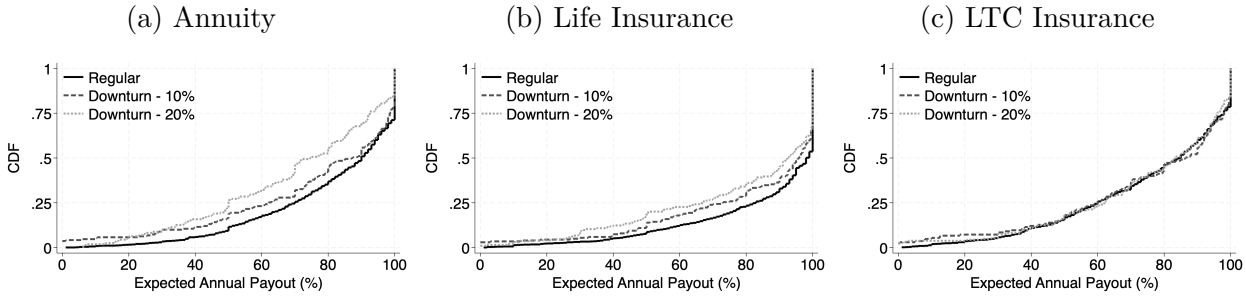


Note: Cross-sectional CDFs of the standard deviation of the subjective payout distribution conditional on a qualifying event. Payouts are expressed as a percentage of the contracted payout.

B.1 Payout Beliefs Conditional on Aggregate Financial Conditions

To examine the drivers of risk perceptions, Figure B.2 presents subjective annual payout distributions conditional on different aggregate financial conditions, proxied by stock market returns. Relative to the baseline, we find that respondents expect annuity and life insurance payments to decrease in economic downturns. The payout CDFs shift leftward conditional on a 10% market decline, with a more pronounced shift at 20%. In contrast, the subjective LTC insurance payout distribution is invariant to market conditions. These divergent patterns suggest that perceived sources of risk vary by insurance type. LTCI risk may be associated with claim denials stemming from subjective state verification by insurance companies, whereas annuity and life insurance risks may be associated with the solvency of insurers. These patterns also indicate a degree of sophistication in survey responses and nonpayment beliefs.

Figure B.2: **Expected Payout Conditional on Aggregate Financial Conditions**



Note: Cross-sectional CDFs of expected mean payout shares conditional on aggregate financial conditions, proxied by annual stock market returns. Beliefs are reported for "regular" times and for 10% and 20% annual stock market declines.

Figure B.3: **Factors Considered when Responding to the Survey**



Note: Fraction of respondents citing each factor as a substantive consideration in their nonpayment beliefs.

B.2 Survey Response Quality and Credibility

The validity of our findings relies on the credibility of the elicited subjective beliefs. First, the survey design allows us to rule out some framing effects. Randomizing module order and hypothetical payout amounts yields no significant differences in the core estimates of Table 3. Second, we verify internal consistency using a qualitative ranking of insurance riskiness as described in Section 2.3.2. We classify 84% of respondents as internally consistent, meaning that their qualitative ranking of the products aligns with their quantitative responses. Table B.2 shows that our results are robust to restricting the analysis to this subsample.

In addition, several features of survey responses provide evidence of answer quality. For example, there is a consistent ordinal ranking of the perceived riskiness of insurance products across individuals. More importantly, the ability of these subjective measures to predict actual insurance ownership establishes their external validity, directly linking elicited beliefs to observed behavior. Similarly, the elicited conditional expectations reveal that respondents

associate aggregate financial risk with annuity and life insurance payout risk, but not with LTCI risk.

Finally, self-evaluation metrics indicate a high degree of confidence in individuals' responses. As reported in Figure B.3, respondents considered a number of reasonable factors when formulating their answers. Moreover, only 17% of the sample reported not being confident in their answers, and only 9% indicated having given no prior thought to the core concerns of the survey. Overall, these patterns suggest that the survey captures considered, economically meaningful, subjective beliefs.

Table B.1: **Regression of Ownership on Risk Metrics**

	Own Annuity	Own Life	Own LTCI	Own Annuity	Own Life	Own LTCI
Payment Exp. Val ($E[\omega^k]$)	-0.0018	0.0048***	0.0007	-0.0002	0.0040**	0.0007
	(0.194)	(0.001)	(0.090)	(0.770)	(0.010)	(0.131)
Full Def. Prob (δ^k)	-0.0022***	-0.0015	-0.0023***	-0.0021***	-0.0012	-0.0022***
	(0.000)	(0.129)	(0.000)	(0.000)	(0.167)	(0.000)
Payment S.D. ($\sigma(\omega^k)$)	-0.0043**	-0.0006	-0.0008	-0.0025***	-0.0006	-0.0008
	(0.002)	(0.712)	(0.246)	(0.001)	(0.733)	(0.246)
Trust				0.0194	-0.0053	0.0165
				(0.080)	(0.795)	(0.231)
Cognitive Score				-0.0011	-0.0023	0.0005
				(0.584)	(0.447)	(0.836)
Financial Literacy Score				-0.0149	-0.0696*	-0.0155
				(0.331)	(0.012)	(0.327)
Numeracy Score				-0.0081	0.0225	-0.0223
				(0.550)	(0.289)	(0.122)
Experienced Fraud				0.0369	0.0211	-0.0136
				(0.437)	(0.743)	(0.746)
Risk Aversion				-0.0069	-0.0156	-0.0020
				(0.277)	(0.085)	(0.699)
Propensity to Plan				0.0140	0.0014	0.0030
				(0.233)	(0.944)	(0.792)
Early Stock Returns				0.4653	-0.4527	-0.1782
				(0.140)	(0.363)	(0.629)
Male	0.0528*	-0.0213	-0.0208	0.0479	-0.0272	0.0066
	(0.047)	(0.619)	(0.445)	(0.100)	(0.543)	(0.799)
Age	-0.0074	0.0238	-0.0191	0.0126	0.0039	-0.0297
	(0.641)	(0.304)	(0.271)	(0.427)	(0.897)	(0.229)
Higher Education	0.0447	0.1087	0.1029	0.0539	0.1243	0.1039
	(0.347)	(0.394)	(0.097)	(0.318)	(0.280)	(0.163)
Married	-0.0327	-0.0484	0.0478	-0.0253	-0.0574	0.0404
	(0.259)	(0.401)	(0.220)	(0.393)	(0.313)	(0.299)
Retired	0.0212	-0.0693	0.0576	0.0023	-0.0966	0.0499
	(0.638)	(0.203)	(0.088)	(0.956)	(0.063)	(0.142)
Bad Health	-0.0706	-0.0450	-0.0394	-0.0539	-0.0515	-0.0442
	(0.053)	(0.484)	(0.076)	(0.116)	(0.408)	(0.088)
LTC Health	-0.0901**	0.0520	0.0285	-0.0963*	0.0601	0.0046
	(0.003)	(0.602)	(0.639)	(0.034)	(0.515)	(0.940)
Income ($million)	-0.0502	1.1503**	0.6993**	-0.0623	1.0854**	0.6390*
	(0.872)	(0.005)	(0.008)	(0.842)	(0.004)	(0.013)
Wealth ($million)	-0.0229*	-0.0041	-0.0003	-0.0195	-0.0042	-0.0013
	(0.049)	(0.982)	(0.986)	(0.095)	(0.814)	(0.948)
N	1054	1045	1039	1054	1045	1039
Pseudo R^2	0.161	0.132	0.129	0.267	0.214	0.175

p-values in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.2: **Regression of Ownership on Risk Metrics: Sample of Internally-Consistent Respondents**

	Own Annuity	Own Life	Own LTCI	Own Annuity	Own Life	Own LTCI
Payment Exp. Val ($E[\omega^k]$)	-0.0019	0.0048***	0.0009*	0.0002	0.0037*	0.0008
	(0.236)	(0.001)	(0.047)	(0.745)	(0.018)	(0.098)
Full Def. Prob (δ^k)	-0.0017**	-0.0015	-0.0022***	-0.0017**	-0.0011	-0.0022***
	(0.007)	(0.129)	(0.000)	(0.004)	(0.228)	(0.000)
Payment S.D. ($\sigma(\omega^k)$)	-0.0046**	-0.0006	-0.0009	-0.0024**	-0.0008	-0.0008
	(0.003)	(0.712)	(0.257)	(0.004)	(0.656)	(0.308)
Trust				0.0227	-0.0117	0.0233
				(0.057)	(0.597)	(0.123)
Cognitive Score				-0.0018	-0.0022	0.0015
				(0.438)	(0.502)	(0.581)
Financial Literacy Score				-0.0041	-0.0793**	-0.0106
				(0.808)	(0.009)	(0.574)
Numeracy Score				-0.0205	0.0240	-0.0312
				(0.174)	(0.305)	(0.064)
Experienced Fraud				0.0549	0.0459	0.0002
				(0.319)	(0.514)	(0.996)
Risk Aversion				-0.0086	-0.0142	-0.0026
				(0.200)	(0.154)	(0.649)
Propensity to Plan				0.0018	-0.0131	0.0063
				(0.882)	(0.527)	(0.624)
Early Stock Returns				0.3934	-0.8636	-0.1302
				(0.231)	(0.088)	(0.760)
Male	0.0516	-0.0213	-0.0338	0.0503	-0.0461	0.0030
	(0.086)	(0.619)	(0.269)	(0.123)	(0.350)	(0.922)
Age	0.0049	0.0238	-0.0325	0.0218	-0.0110	-0.0423
	(0.752)	(0.304)	(0.089)	(0.204)	(0.720)	(0.124)
Higher Education	0.0280	0.1087	0.1145	0.0442	0.2009	0.1160
	(0.561)	(0.394)	(0.104)	(0.429)	(0.109)	(0.167)
Married	-0.0312	-0.0484	0.0619	-0.0229	-0.0562	0.0543
	(0.330)	(0.401)	(0.154)	(0.479)	(0.352)	(0.205)
Retired	0.0516	-0.0693	0.0651	0.0239	-0.1351*	0.0636
	(0.310)	(0.203)	(0.084)	(0.599)	(0.014)	(0.097)
Bad Health	-0.0532	-0.0450	-0.0314	-0.0373	-0.0380	-0.0356
	(0.177)	(0.484)	(0.212)	(0.299)	(0.575)	(0.224)
LTC Health	-0.1055***	0.0520	0.0393	-0.1258*	0.0815	0.0066
	(0.001)	(0.602)	(0.566)	(0.012)	(0.406)	(0.925)
Income ($million)	0.3509	1.6675**	0.7293*	0.3376	1.4445***	0.6152*
	(0.275)	(0.005)	(0.027)	(0.295)	(0.001)	(0.048)
Wealth ($million)	-0.0121	-0.0118	0.0066	-0.0092	0.0006	0.0066
	(0.335)	(0.982)	(0.769)	(0.466)	(0.974)	(0.762)
N	870	861	855	870	861	855
Pseudo R^2	0.136	0.132	0.146	0.249	0.233	0.195

p-values in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

C Objective Evidence on Insurance Nonpayment

Guaranty protection and insurer resolutions. Typical statutory limits are approximately $300,000 for life-insurance death benefits, $250,000 in annuity present value, and $300,000 in long-term-care benefits, although limits and procedures vary across states (National Organization of Life and Health Insurance Guaranty Associations, 2025). For a respondent at the sample mean age of 61, the survey's reference annuity of $20,000 per year is worth roughly $270,000 to $310,000 at the 2.5 percent pricing rate of Section 4.3 and the model's health transitions, above the annuity limit. Penn Treaty spent eight years in rehabilitation before its 2017 liquidation. The state estimated that about half of the 76,000 policyholders of Penn Treaty and its affiliate would have claims above their guaranty coverage (Pennsylvania Insurance Department, 2017; Hartley, 2024). Policyholders of Executive Life, the largest failure the guaranty system has handled, had recovered an estimated 86 percent of their expected account values as of 2005 by the state auditor's count (California State Auditor, 2008; Hartley, 2024). About 1,500 structured-settlement annuitants of its New York affiliate had their benefits reduced in its 2012 liquidation (Hartley, 2024).

Claim outcomes reported to state regulators. Insurers file the Market Conduct Annual Statement (MCAS) with each participating state under line-specific filing requirements. The NAIC publishes the jurisdiction-wide ratios for each data year since 2021 as State Ratio Distribution Reports (National Association of Insurance Commissioners, 2026). Table C.1 reports, for each ratio, the median across states of the jurisdiction-wide ratio, computed as the sum of numerators over the sum of denominators across all filers in the state. A claimant request is the initial request for a benefit-eligibility determination, and a benefit payment request is each subsequent request for payment. A request is counted as denied when it is "not paid for any reason," with reason codes for claims the claimant did not pursue, preexisting conditions, the elimination period, services not covered, providers not qualified, and eligibility criteria not met. For life insurance, "denied, resisted, or compromised" includes claims contested within the contestability period. Ratios are suppressed where three or fewer companies report, company-level submissions are not published, and annuity filings contain no claim fields.

Table C.1: **Claim Outcomes in NAIC Market Conduct Annual Statements**

	2021	2022	2023	2024	2025
Long-term care insurance (stand-alone)					
Claimant-request denial ratio	23.0	22.4	21.2	20.8	20.0
Eligibility determinations taking more than 60 days	23.3	24.5	23.4	21.7	21.4
Benefit-payment denial ratio	6.5	6.6	7.4	8.1	8.1
Life insurance, death claims					
Denied, resisted, or compromised, non-cash-value policies	1.8	2.8	2.6	2.1	2.0
Denied, resisted, or compromised, cash-value policies	0.6	0.6	0.7	0.6	0.6
Paid more than 60 days after proof of loss, non-cash-value	2.0	2.9	2.3	1.7	2.0
Paid more than 60 days after proof of loss, cash-value	2.3	3.9	2.3	2.3	1.8
Disability income insurance					
Claims denied	24.3	24.4	24.2	25.3	25.6

Note: Percent. Median across 49 to 52 jurisdictions of the jurisdiction-wide ratio in the NAIC MCAS state ratio distribution reports (National Association of Insurance Commissioners, 2026). A claimant request is the initial request for a benefit-eligibility determination, and a benefit payment request is each subsequent request for payment. The LTC claimant-request denial ratio divides requests denied or not paid by new claimants plus requests pending at the start of the year; the benefit-payment denial ratio divides payment requests denied or not paid by requests received plus requests pending; the life ratios divide by claims closed or claims paid. Denials are counted for any reason, including requests the claimant did not pursue.

D Estimated Inputs for Model

This appendix describes the health transitions, medical cost shocks, and income profiles that serve as inputs to the structural model.

D.1 Health Transitions and Cost Shocks

Health transitions are estimated using the RAND HRS waves 2 to 12. We construct the defined health states $s \in \{0, 1, 2\}$ from two sets of questions. The first makes use of self-reported subjective health-status questions to classify respondents into good ($s = 0$) or bad ($s = 1$) health.[25] The second set of questions is used to determine whether an individual is in the LTC ($s = 2$) health state. This set of questions presents five ADLs and asks whether respondents receive help with any of the five activities. If the respondent answers yes to at least two of these questions, then we define that respondent to be in the LTC health state.

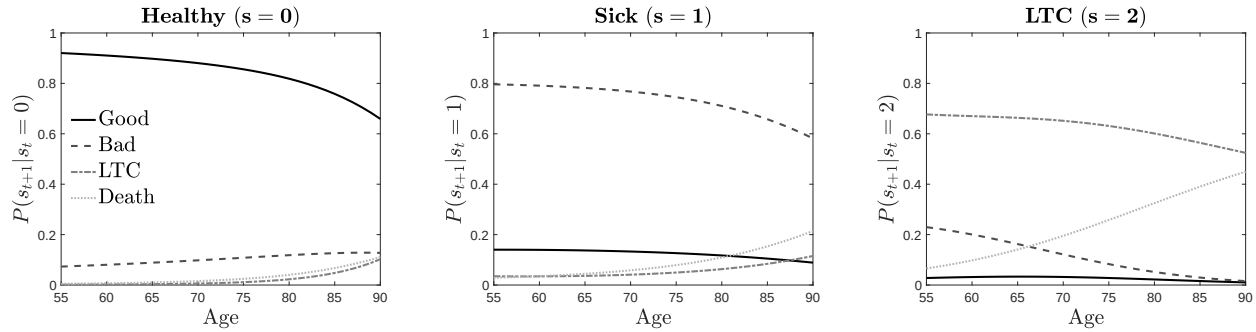
We then estimate a health transition matrix that is conditional on the individual's age, t, sex, f, and permanent income quintile (PIQ), y, constructed in Appendix D.2. In particular, we allow for the health transition to differ between those with lower permanent income, $y \in \{1, 2, 3\}$, and those with higher permanent income, $y \in \{4, 5\}$. This is to allow the model to capture the well-documented wealth gradient in health and longevity.

Figures D.1 and D.2 display the estimated health transitions. Those with higher permanent income have more favorable health transitions. For example, they are more likely to stay healthy throughout their lives and to survive conditional on requiring LTC.

Health Cost Shocks. To estimate the medical cost distribution, we define expenses as out-of-pocket medical expenses using the variable r12oopmd in the RAND HRS in 2014. We subtract out-of-pocket home health and nursing home payments, because these expenses are specific to the LTC state, which the model treats separately. We restrict the 2014 sample to single respondents aged 55 to 100. We estimate the distribution following the process outlined in Ameriks et al. (2020). To estimate the mean of the medical cost distribution, we regress the log of medical expenses on a quartic polynomial in age, health status, sex and health interacted with age. Using the residuals from this first regression, we regress the squared residuals on the same set of state variables as in the first regression to find the conditional variance of medical expenses. The resulting mean and variance estimates parameterize the lognormal distribution for medical expenses conditional on age, sex, and health status. We

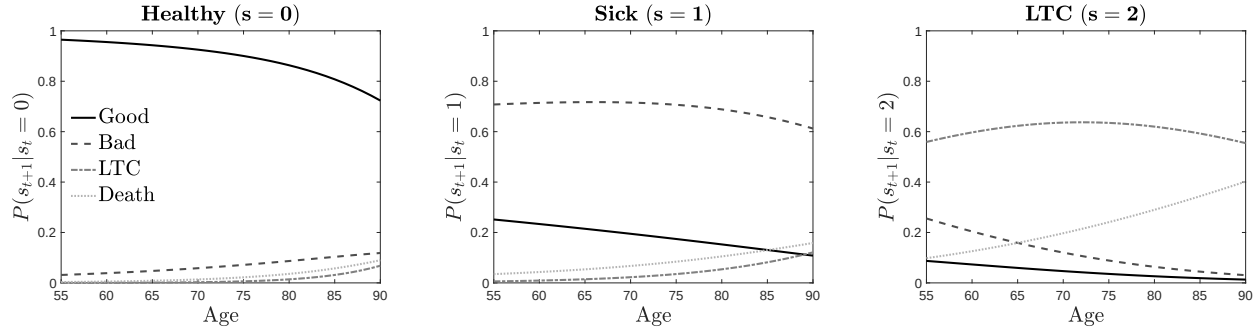
[25] Individuals are defined as in good health if they report health being good, very good, or excellent and are defined to be in bad health if they report health being poor or fair. Self-reported assessments of health status have strong predictive power for future health realizations.

Figure D.1: **Health Transition Profiles: Lower Income** ($PIQ \in \{1, 2, 3\}$)



Note: Each panel reports model-implied age-specific health transition probabilities for individuals in the bottom 3 permanent income quintiles. The three panels correspond to current health states good, bad, and LTC. Within each panel, the four lines show the probability of transitioning to good health, bad health, LTC, or death.

Figure D.2: **Health Transition Profiles: Higher Income** ($PIQ \in \{4, 5\}$)



Note: Each panel reports model-implied age-specific health transition probabilities for individuals in the top two permanent income quintiles. The three panels correspond to current health states good, bad, and LTC. Within each panel, the four lines show the probability of transitioning to good health, bad health, LTC, or death.

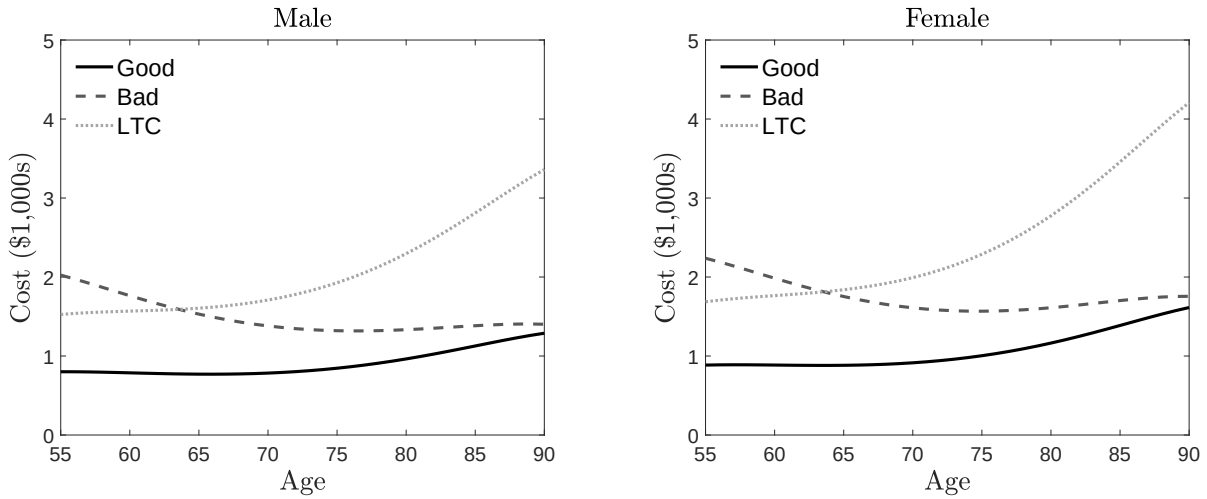
then discretize the error term, using the Tauchen method to generate the medical expense process.

Figure D.3 plots the mean medical cost shock over the life cycle for both men and women. Men in poor health at age 65 spend about $750 more on average than men in good health. At older ages, men in the LTC state spend about $2,000 more per year than men in good health, excluding nursing home and home health services.

D.2 Income Process

We estimate a deterministic income profile for five separate income quintiles. We define income using the RAND HRS 2014 dataset as the sum of income from earnings, public and private pensions and annuities, Social Security income, disability and retirement, unemployment

Figure D.3: **Mean Health Cost Shock Profile**



Note: The figure plots the mean of the medical cost shock distribution over the life cycle for males (left panel) and females (right panel).

compensation, and other government transfers.[26] We evaluate this at the household level by summing up the income of both the respondent and the spouse. We take the income for each household every time we observe them between wave 2 (1994) and wave 12 (2014) of the RAND HRS dataset. Income is evaluated at constant 2014 dollars.

We first adjust income for taxes, because progressive taxes compress differences in disposable income across quintiles and cohorts. We use the federal tax function of Borella, De Nardi and Yang (2023),

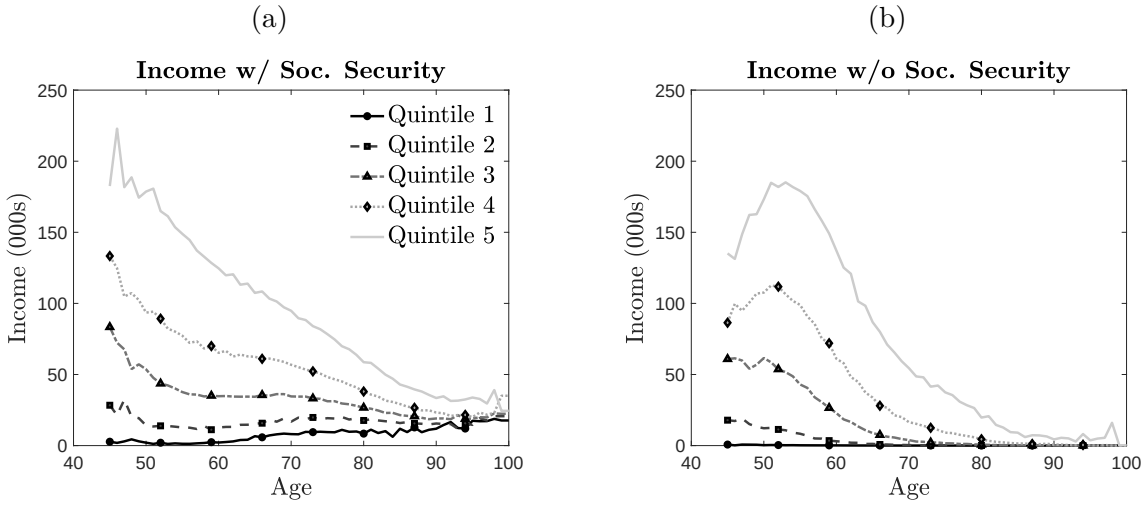
$$T(Y) = Y - (1 - \lambda_T)Y^{1-\tau},$$

with their year-specific estimates of (λ_T, τ). They estimate these parameters from PSID data with tax payments imputed using NBER TAXSIM. The function excludes state income taxes. Fleck, Heathcote, Storesletten and Violante (2025) show that state taxes affect the level but not the progressivity of taxes on average, so federal taxes capture the progressivity that matters for the income profiles.

To assign households to permanent income quintiles, we split the sample into 5-year age brackets (45–49, 50–54, and so on) and, in each wave, place each household in an income quintile within its age bracket, using the 20th, 40th, 60th, and 80th percentiles as thresholds. A household's permanent income quintile is its average quintile across the waves in which we observe it, rounded to the nearest integer.

[26]More specifically, our measure of income is the sum of Respondent and spouse earnings (RwIEARN, SwIEARN), pensions and annuities (RwIPENA, SwIPENA), SSI and Social Security Disability (RwISSDI, SwISSDI), Social Security retirement (RwISRET, SwISRET), unemployment and workers compensation (RwIUNWC, SwIUNWC) and other government transfers (RwIGXFR, SwIGXFR).

Figure D.4: **Deterministic Income Profiles by Quintile**



Note: Each line reflects the age profile of deterministic income for a given income quintile, estimated from the HRS panel data. The left panel includes Social Security payments, while the right panel excludes them.

We then run the following panel regression across all waves w:

$$\begin{aligned}\log(inc_{i,w}) = \alpha_i + \beta_1 Age_{i,w} + \beta_2 Age_{i,w}^2 + \beta_3 Age_{i,w}^3 + \beta_4 Age_{i,w}^4 \\ + \beta_5(Male_i \times Age_{i,w}) + \beta_6(PIQ_i \times Age_{i,w}) + \epsilon_{i,w}\end{aligned}$$

where $inc_{i,w}$ is the income of household i in wave w. The household fixed effects α_i control for cohort effects that would otherwise confound the age profile. If aggregate real income grows over time, for example, a regression without them would overstate the decline of income with age.

For each age $t \in \{45, \ldots, 99\}$, quintile $PIQ \in \{1, \ldots, 5\}$, and sex f, the deterministic income profile is the regression's prediction at (t, PIQ, f), with α_i replaced by the average fixed effect of households in that cell.[27]

The profiles that we feed into the model for each quintile are shown on the left-hand side of Figure D.4. The profiles differ substantially across quintiles. All of them decline in later years.

For the Social Security counterfactual of Section 5.6, we repeat these steps with income that excludes Social Security benefits. The right panel of Figure D.4 shows the resulting profiles. They lie below the baseline profiles, proportionally more so for the lower quintiles.

[27]Results are very similar, although a bit smoother, if we do not condition on age t in the cell.

D.3 Assets

Financial wealth is defined as the sum of an IRA (individual retirement account) and employer-sponsored retirement, checking, saving, money market, mutual fund, certificate of deposit, brokerage, and education-related accounts. Total wealth is defined as financial wealth plus housing wealth. We set the locked-in annual premia on respondents' existing LTC and life insurance policies to the prices they would have paid at age 45 in good health.

E Simulation and Calibration of Model

This appendix describes how we simulate the model and how we use the simulations to calibrate it to moments of wealth, insurance ownership, and SSQ responses from Ameriks et al. (2020).

E.1 Model Simulations

We first solve the model to compute optimal decision rules for wealth, insurance ownership, consumption and government care take-up, given values of the calibrated parameters Θ (Appendix E.2). The rules are functions of the individual state variables $S_i = \{s_i, f, y_i, b_i, x_i^{\text{Ann}}, x_i^{\text{LI}}, x_i^{\text{LTC}}, p_i^{\text{LTC}}, p_i^{\text{LI}}\}$. The model is solved by conducting a grid search over the state variables. Once we have the policy functions defined over S_i, we then simulate the model for all 1,085 individuals i in our UAS sample. To set the initial conditions, we take from the UAS sample the empirical counterparts of the state variables, $\hat{S}_i$ for each individual. We simulate each individual 100 times, drawing relevant shocks $\hat{\Psi}$ from the distributions of full default, annual nonpayment, medical costs as well as the health transitions, and simulate the behavior implied by the optimal policies.[28] We then aggregate these individual behaviors to construct simulated moments from the model, $o(\hat{S}, \Theta, \hat{\Psi})$, which we compare with the empirical targets. Each simulated individual keeps respondent i's UAS weight throughout the simulation. For example, the life insurance participation rate at age 57 is the weighted average over all simulated observations at age 57.

Simulating this model is computationally intensive. First, solving the policy functions from ages 45 to 99 via backward induction requires a fine grid search across nine dimensions. Then, the model must be simulated 100 times for each of the 1,085 individuals. Solving the model for a single parameter set Θ takes approximately 36 hours. To manage this, we use more than 200 computing nodes, enabling extensive parallelization. This setup significantly reduces

[28] Results are essentially equivalent if we simulate each individual just 10 times.

computation time and allows us to evaluate policy functions across many combinations of Θ when calibrating the model.

E.2 Calibration Strategy

We develop a calibration strategy similar to those used in De Nardi et al. (2010), Gourinchas and Parker (2002) and French and Jones (2011) to calibrate the parameter set $\Theta := \{\beta, \theta_{3,45}, \gamma, \rho, r\}$. All other parameters of the model, outlined in Tables 5 and 6, are externally calibrated without the use of the structural model. There are three sets of moments that we use to calibrate the model. First, we target wealth percentiles across the life cycle, which are commonly used in life-cycle savings models. Second, we target average participation rates for each of the three insurance products that we analyze. Finally, we target a set of moments derived from SSQ responses drawn from Ameriks et al. (2020).

Let $Z = (Z_i)_{i=1}^{I}$ collect the measurements for all individuals, including wealth and insurance holdings, SSQ responses, and state variables. Let $g(\hat{S}, \Theta, \hat{\Psi}) := o(\hat{S}, \Theta, \hat{\Psi}) - s(Z)$ be the vector of differences between the simulated statistics and their empirical counterparts, $s(Z)$. We solve

$$\hat{\Theta} = \arg\min_{\Theta} g(\hat{S}, \Theta, \hat{\Psi})' W g(\hat{S}, \Theta, \hat{\Psi})$$

where W is the weighting matrix. Moments within each of the three sets receive equal weights, and the three sets receive equal total weight.

E.2.1 Wealth Moments

The first set of moments consists of age-specific wealth percentiles. Specifically, we target the 25th, 50th, and 75th percentiles of the wealth distribution within 5-year age brackets (50–54, 55–59, . . . , 80–84). We stop at age 85 due to small sample sizes at older ages. This yields seven age groups and, with three percentiles per group, a total of 21 targeted wealth moments. Model-implied percentiles are directly compared to their empirical counterparts.

To construct the wealth moments in the model, we treat each simulated individual-age observation as a cross-sectional data point. That is, every time one of the 108,500 simulated individuals is observed at a given age, it enters the corresponding age bracket. For example, the percentiles for ages 50–54 are computed using all simulated observations that fall within that age range.

Empirical wealth moments are constructed using the HRS. We use the HRS rather than the UAS because it provides a larger and therefore less noisy estimate of the wealth distribution.

This approach relies on the UAS being broadly representative of the HRS population. As shown in Table 1, however, the UAS sample differs somewhat from the HRS, particularly in marital and retirement status.

To facilitate comparisons across the two surveys, we harmonize them along these two dimensions. Consider, for example, the cohort aged 60–64. Let (m^H_{RET}, m^H_{MAR}) denote the fractions of individuals in this cohort who are retired and married, respectively, in the HRS, and let (m^U_{RET}, m^U_{MAR}) denote the corresponding fractions in the UAS. We then adjust the HRS sampling weights as follows:

1. (Retired, Married): scale by $\frac{m^U_{RET} m^U_{MAR}}{m^H_{RET} m^H_{MAR}}$
2. (Retired, Not Married): scale by $\frac{m^U_{RET}(1-m^U_{MAR})}{m^H_{RET}(1-m^H_{MAR})}$
3. (Not Retired, Married): scale by $\frac{(1-m^U_{RET})m^U_{MAR}}{(1-m^H_{RET})m^H_{MAR}}$
4. (Not Retired, Not Married): scale by $\frac{(1-m^U_{RET})(1-m^U_{MAR})}{(1-m^H_{RET})(1-m^H_{MAR})}$

We reweight the HRS sample so that its composition along these dimensions matches that of the UAS. Figure 5 presents the resulting comparison. Reweighting lowers average HRS wealth to \$526,000, closer to the UAS average of \$461,000 (Table 1).

E.2.2 Insurance Participation Rates

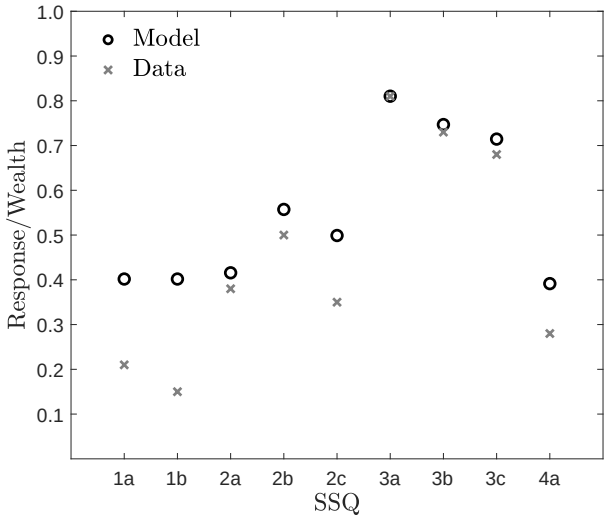
We compute insurance participation rates in the same way as the wealth moments. The key difference is that we only target the aggregate participation rates of life insurance, annuities and LTCI, leaving untargeted the life-cycle profile of insurance ownership. Our empirical moments are the aggregate participation rates from the 2014 HRS, using the adjusted sample weights. This generates empirical targets of 61% for life insurance, 12% for LTCI, and 4% for annuities.

E.2.3 SSQ Moments

This final moment set is constructed entirely from SSQ responses of the sample of individuals in Ameriks et al. (2020). The key difference here is that we fix the parameter values of $(\theta_2, \bar{\theta}_3, \kappa_2, \kappa_3, \bar{c}_2)$ to those estimated by the paper as their posterior means, leaving us only with γ as the calibrated variable. The intuition for this is that γ is the parameter that, locally, most affects insurance ownership moments, which were not studied in Ameriks et al. (2020).

As moments, we target the model-implied responses to the empirical mean of each of the nine SSQ variants, as a fraction of the maximum response. Figure E.1 compares model and data.

Figure E.1: **Mean SSQ Responses in Model (circles) and Data (crosses)**



Note: Mean responses to the nine SSQs of Ameriks et al. (2020). To match low insurance demand, our calibration has lower relative risk aversion than Ameriks et al. (2020). This worsens the fit to questions 1a and 1b.

E.3 Computing Consumption-Equivalent Welfare

For scenario $S \in \{A, B\}$ and individual i, let the (simulated) lifetime utility be

$$\widehat{V}_i(S) \equiv \frac{1}{N} \sum_{n=1}^{N} \left[\sum_{t=0}^{T} \beta^t \, u_{s_{int}} \left(c_{int}^S \right) \right],$$

where n indexes each simulation of health states, medical cost shocks, and insurance default outcomes, c_{int}^S is realized expenditure in health state s_{int}, and s_{int} is the health state of individual i in simulation n at age t. The consumption-equivalent welfare gain of B relative to A for i is the unique $\Delta_i^C \in \mathbb{R}$ that solves

$$\widehat{V}_i(A) \;=\; \frac{1}{N} \sum_{n=1}^{N} \left[\sum_{t=0}^{T} \beta^t \, u_{s_{int}} \left(\left(1 - \Delta_i^C \right) \, c_{int}^B \right) \right].$$

Once we compute Δ_i^C for each individual i, we then report the population consumption-equivalent welfare as the UAS-weighted mean $\Delta^C \equiv \sum_i w_i^{UAS} \Delta_i^C / \sum_i w_i^{UAS}$.

Throughout the paper, we compute welfare effects across various counterfactual scenarios using consumption-equivalent welfare as our evaluation metric. Consider two scenarios, indexed by $\{A, B\}$, where A is the reference scenario and B is the considered scenario. For each individual i, we calculate the percentage reduction in expenditure in every health state,

denoted by Δ_i^C, that the individual would be willing to accept in order to be indifferent between scenarios B and A. To do this, we scale expenditure in every health state by $(1-\Delta_i^C)$ across all simulations in scenario B, then compute the average lifetime utility across those simulations. We choose Δ_i^C such that this average lifetime utility in scenario B equals the average lifetime utility of the same individual across simulations in scenario A. Finally, we aggregate the resulting welfare effects using the UAS weights assigned to each individual.

E.4 Model Sensitivity to Parameters

In conducting the calibration exercise, we learn the sensitivity of our model moments to the parameter set Θ that we are calibrating.

E.4.1 Discount Factor (β)

The discount factor β mainly affects the wealth moments. A higher β raises saving and slows the decline of wealth late in life. A higher β also lowers life insurance holdings. Because the calibrated bequest motive is strongest early in life, a higher β shifts weight toward later years, when life insurance is less valuable. It also slightly reduces LTCI demand.

E.4.2 Risk Aversion (γ)

The risk aversion coefficient γ mainly affects insurance ownership rather than the wealth profiles. A higher γ strongly raises LTCI and life insurance ownership, because both insure unlikely events (needing long-term care or dying young).

Matching the LTCI participation rate requires a relatively low γ. A more risk-averse individual places more value on avoiding the LTC state ($s = 2$) without private insurance, since that state likely means government care, which yields low utility.

E.4.3 Bequest Motive When Young ($\theta_{3,45}$)

This controls the bequest motive of young individuals. Raising $\theta_{3,45}^{-\gamma}$ strengthens the bequest motive of young individuals and raises their life insurance participation. It also raises LTCI ownership at young ages. Entering the LTC state when young sharply raises the probability of dying (Figure D.1), so LTCI provides incidental bequests much as life insurance does.

E.4.4 Bequest Life-Cycle Shape (ρ)

This parameter ρ controls the shape of the bequest motive over the life cycle, governed by equation (2). For ρ that is very close to 0, it approximates a linear function, but as ρ increases

it tilts the bequest motive when young higher and creates a steeper declining slope for later ages. With a linear bequest motive ($\rho \to 0$), life insurance ownership would fall faster with age than in the data.

A much higher ρ would raise life insurance ownership at young ages. It could also make life insurance ownership rise between ages 45 and 60, because the bequest motive would stay strong as the probability of death rises.

E.4.5 Household and Insurer Returns (r and r_I)

The two rates affect the model through different margins. Holding r_I fixed, a higher household return r increases desired wealth accumulation and makes liquid self-insurance more attractive relative to insurance. This reduces insurance demand, particularly for annuities, which require an up-front commitment of wealth. Holding r fixed, a higher insurer pricing rate r_I lowers the present value of promised insurance payouts and therefore lowers contract prices. The effect is strongest for long-duration annuity payments. Raising the two rates together makes saving and long-duration contracting more attractive in absolute terms while leaving much of their relative-return difference unchanged. The wealth moments are especially sensitive to r, whereas the gap $r - r_I$ helps determine the allocation between bonds and insurance.

F Alternative Results: Non-Rational Expectations

This appendix reports selected results from the main-text experiments, but under the alternative assumption that beliefs about nonpayment risk are non-rational. In these experiments, insurance is safe, meaning that payouts are always made. Subjective beliefs, however, are the same as in the main text, calibrated to those elicited in the survey.

F.1 Insurance Ownership and Savings: Non-Rational Expectations

Table F.1 repeats the exercise of Table 8 under non-rational expectations. The results are very similar, because portfolio choice depends on beliefs, whether or not they are accurate. The small differences arise because individuals in this setting receive more payments than they expect. This raises average wealth in the simulations.

F.2 Welfare Effects by Product: Non-Rational Expectations

Table F.1: **Insurance Ownership and Savings: Impact of Risky Insurance and Loads (Non-Rational Expectations)**

	Data	Baseline	Safe Insurance	Zero Loads	Zero Loads & Safe Insurance
A. *Ownership Rate*					
LI	61%	56%	73%	72%	84%
LTCI	12%	15%	43%	30%	55%
Ann	4%	2%	8%	3%	10%
B. Δ *Personal Savings*					
25th percentile	–	−38%	−48%	−47%	−55%
50th percentile	–	−17%	−27%	−23%	−33%
75th percentile	–	−6%	−16%	−14%	−22%

Note: Panel A reports aggregate ownership rates for three insurance products under alternative scenarios. Panel B reports percentage changes in wealth across the distribution, relative to values from a no-insurance benchmark, averaged across age with population weights. Baseline is the calibrated model; Safe Insurance sets default probabilities to zero (i.e., no nonpayment risk); Zero Loads sets product-level loads λ^k to zero; Zero Loads & Safe Insurance combines both adjustments under non-rational expectations.

Table F.2 repeats the same exercise as in Table 10 but under non-rational expectations. The conclusions are the same as under rational expectations. The product-level gains sum approximately to the total, so the products are neither strong substitutes nor strong complements.

Table F.2: **Welfare Gains from Individual Insurance Products (Non-Rational)**

	Baseline	Safe Insurance	Zero Loads	Zero Loads & Safe Insurance
LI	0.7%	0.7%	1.7%	1.7%
LTCI	0.9%	1.3%	1.3%	1.5%
Ann	0.0%	0.2%	0.4%	0.7%
All	1.7%	2.3%	3.4%	4.1%

Note: Consumption-equivalent welfare gains relative to a no-insurance benchmark (non-rational beliefs case). Each row isolates the introduction of only the specified product. The final row introduces all three products and reproduces Panel B from Table 9. Baseline is the calibrated model; Safe Insurance sets all default probabilities to zero; Zero Loads sets product-level loads λ^k to zero; Zero Loads & Safe Insurance sets all nonpayment probabilities and loads to zero.

F.3 Role of Public Long-Term Care: Non-Rational Expectations

Table F.3 runs the same experiment as in Table 11 but under non-rational expectations. Ownership is nearly identical to the rational-expectations case, since portfolio choice depends on beliefs. Welfare losses again rise as the public LTC floor falls and again fall most heavily on poorer individuals, but the losses are smaller: an aggregate loss of 2.8% rather than 4.0% when $\bar{c}_2 = 40$. As the floor falls, individuals buy more LTCI while expecting only partial payouts. Because the contracts pay in full, the unexpected payouts offset part of the welfare loss.

Table F.3: **The Effects of Government LTC Provision (Medicaid): Non-Rational**

	Baseline $\bar{c}_2 = 77.43$	$\bar{c}_2 = 60$	$\bar{c}_2 = 40$
A. *Ownership Rate*			
LI	56%	54%	51%
LTCI	15%	30%	54%
Ann	2%	3%	3%
B. *LTCI Ownership Rate*			
0–25th Wealth Percentile	9%	12%	42%
25–50th Wealth Percentile	28%	44%	73%
50–75th Wealth Percentile	15%	30%	56%
75–100th Wealth Percentile	12%	36%	46%
C. *Welfare Effects*			
Aggregate	-	-1.0%	-2.8%
0–25th Wealth Percentile	-	-1.2%	-4.6%
25–50th Wealth Percentile	-	-1.4%	-4.7%
50–75th Wealth Percentile	-	-1.1%	-1.6%
75–100th Wealth Percentile	-	-0.3%	-0.3%

Note: Columns correspond to alternative levels of the public long-term care consumption guarantee ($\bar{c}_2$). Panel A reports model-implied insurance participation rates. Panel B reports private long-term care insurance ownership rates across the wealth distribution. Panel C reports consumption-equivalent welfare changes relative to the Baseline, in aggregate and by wealth percentile, using population weights.

F.4 Role of Social Security: Non-Rational Expectations

Table F.4 repeats the Social Security counterfactual of Table 12 under non-rational expectations. Participation rates are nearly the same as in the rational-expectations case. The welfare gain from replacing Social Security with a lump-sum transfer is 1.6%, close to the 2.0% gain under rational expectations.

Table F.4: **The Effects of Social Security: Non-Rational Expectations**

	Data	**No Insurance**	**Baseline**	**No Social Security**
A. *Ownership Rate*				
LI	61%	0%	56%	20%
LTCI	12%	0%	15%	14%
Ann	4%	0%	2%	4%
B. *Welfare Gains*				
Δ Welfare	-	-	-	1.6%

Note: Panel A reports insurance ownership rates in the data and across model scenarios. Panel B reports consumption-equivalent welfare changes relative to Baseline. No Social Security removes Social Security payments and rebates their present value to individuals as a lump-sum transfer.